# Is Collaboration Worth It? A Decision-Oriented Survey in Multi-Agent Systems

Guanjie Cheng[1], Yifan Zhang[1], Zewen Lin[1], Xinkui Zhao[1], Sai Liu[1], Jianwei Yin[2], and Shuiguang Deng[2]

[1]School of Software Technology, Zhejiang University
[2]The College of Computer Science and Technology, Zhejiang University

January 30, 2026

Posted on 30 Jan 2026 — CC-BY 4.0 — https://doi.org/10.36227/techrxiv.176978654.41579706/v1 — e-Prints posted on TechRxiv are preliminary reports that are not peer reviewed. They should not b...

# Is Collaboration Worth It? A Decision-Oriented Survey in Multi-Agent Systems

**Guanjie Cheng**[1] **Yifan Zhang**[1,*] **Zewen Lin**[1] **Xinkui Zhao**[1,*] **Sai Liu**[1]
**Jianwei Yin**[2] **Shuiguang Deng**[2]

[1]School of Software Technology, Zhejiang University
[2]The College of Computer Science and Technology, Zhejiang University

***Abstract*—Multi-agent collaboration now underpins critical systems—from warehouse swarms to LLM-based agentic workflows—yet collaboration is never free. It competes with hard resource budgets (bandwidth, latency, energy, compute) and soft frictions (partial observability, strategic coupling, and non-stationarity). While prior surveys largely catalog method families, they often fail to address the practitioner's fundamental decision problem: determining when collaboration justifies its cost and how to design mechanisms that survive real-world constraints.**

**We present a decision-oriented survey structured around four pivotal research questions. RQ1 (Communication) investigates the value of interaction under explicit budgets, identifying when shared representations, topology design, or temporal sparsity can substitute for costly messaging. RQ2 (Credit Assignment) focuses on aligning local learning with system-level objectives, synthesizing mechanisms—from counterfactual baselines and Shapley values to structured critics—that ensure faithful attribution under heterogeneity. RQ3 (Scalability) examines the boundaries of collaboration, analyzing saturation effects, cascading failures, and the onset of negative collaboration as populations and task densities scale. RQ4 (Evaluation) proposes decision-aligned protocols that replace single-point metrics with efficiency frontiers and deployability gates, explicitly measuring alignment and trustworthiness alongside capability.**

**Synthesizing empirical regularities and decision rules across these dimensions, we outline a compact reporting standard and future trends toward adaptive orchestration and decentralized ecosystems. Our goal is to transform a fragmented evidence base into defensible guidance for designing and deploying collaborative AI systems at scale.**

***Index Terms*—Multi-Agent Systems (MAS), Multi-Agent Collaboration (MAC), Multi-Agent Communication, MAS Scalability, Agent Evaluation, LLM-enabled Agents**

## I. Introduction

Multi-agent collaboration [1], [2], [3] has moved from a thought experiment in game-theoretic toy worlds to the operating principle behind real systems—robotic swarms coordinating in warehouses [4], [5], [6], fleets of connected vehicles merging at on-ramps [7], [8], [9], and LLM-driven software agents negotiating access to scarce tools and APIs [10], [11], [12]. In each of these settings, collaboration is never free. It competes with hard, measurable constraints—wireless bandwidth [9], [13], energy envelopes [14] and compute quotas [15]—and with softer but equally binding frictions: misaligned local objectives, strategic behavior, and the non-stationarities that emerge as populations scale and tasks drift. A practical survey must therefore start from a decision problem that practitioners actually face: **When is collaboration worth it, for whom, and under which constraints can it be made to scale reliably?**

We take this decision problem as our North Star and organize the survey around the minimal ingredients needed to answer it: the value of communication under resource constraints, the alignment of local learning signals with system-level objectives, and the scaling laws and failure modes that appear as populations grow or migrate to new regimes. The resulting perspective is intentionally decision-oriented: instead of re-listing algorithm families, we synthesize principles, thresholds, and reporting protocols that allow readers to choose and justify a design.

### A. Motivation

Current research often treats multi-agent collaboration as an unbounded optimization problem. However, practical deployment reveals four critical gaps that motivate our study:

- **Collaboration is budgeted, not unlimited.** Real systems operate under strict constraints, including latency, token limits, and reliability SLAs. Excess interaction violates real-time requirements, while insufficient interaction prevents necessary coordination. This frames collaboration as a budget allocation problem, motivating our investigation into efficient communication modes.

* Corresponding author.

- **Local actions do not automatically optimize system welfare.** Agents face partial observability and possess heterogeneous capabilities. Without appropriate coordination, local improvements often diverge from the global objective. This necessitates credit assignment mechanisms that translate global goals into locally actionable signals, ensuring alignment between individual policies and system welfare.
- **Collaboration patterns are fragile at scale.** Methods that succeed in small teams often degrade as populations grow or task densities increase. Phenomena such as communication saturation and cascading failures only emerge at scale. A robust design must therefore identify structural priors that allow systems to scale without collapsing.
- **Evaluation must measure deployability, not just capability.** Standard benchmarks often report success rates under generous assumptions, ignoring the "net value" of collaboration. Real-world adoption requires meeting safety and cost thresholds. This demands decision-aligned evaluation protocols that report performance–cost frontiers rather than single-point metrics.

### B. Gap

Existing surveys have advanced the field by cataloging learning paradigms [1], [16], [2], [3], [17], [18] and by mapping application domains [19], [7], [20], [21]. Yet two critical ingredients for actionable decisions are typically missing.

First, decision rules are under-articulated. We lack **prescriptive** guidance for design trade-offs: When does explicit messaging outperform implicit coordination via shared representations? Under what bandwidth and latency budgets does additional communication cease to be beneficial? Which forms of credit assignment are robust to specific task couplings? As a result, practitioners are often forced to extrapolate from incomparable case studies without a unified framework.

Second, evaluation protocols are seldom aligned with collaboration decisions. Many studies optimize algorithms under generous communication or compute assumptions, reporting only task success or return. Critical deployment metrics—communication load, end-to-end latency, energy, and population-level stability—are inconsistently measured or omitted. Without protocols that bind metrics to decision variables, current benchmarks risk optimizing for **laboratory feasibility rather than real-world deployability**, making evidence difficult to accumulate.

This survey addresses both gaps by (i) elevating key design trade-offs into explicit research questions and (ii) proposing evaluation protocols that report the metrics necessary to answer those questions.

### C. Research questions

We distill the decision problem—*when collaboration is worth it under real deployment constraints*—into four tightly coupled research questions (RQs) that structure the remainder of the paper:

- **RQ1: Communication value under constraints.** Given budgets on bandwidth, latency, energy, and compute, when is communication worth the cost, and what is the minimal sufficient form? We connect marginal performance gains to marginal bits/time/energy, and synthesize design choices spanning explicit messaging, shared representations/distillation, topology design, and temporally sparse protocols.
- **RQ2: Credit Assignment: Aligning Agents with System.** In collaborative learning, how can we construct agent-wise learning signals that faithfully reflect each agent's contribution to a system-level objective under partial observability, heterogeneous costs, and strategic coupling? We review credit assignment as contribution attribution for learning (e.g., counterfactual baselines, difference rewards, Shapley-style signals), structured critics and value/policy decompositions, and stabilized training protocols that reduce variance, mitigate mis-crediting, and support scalable decentralized execution.
- **RQ3: Scalability: From Small Groups to Complex Dimensions.** How do collaboration patterns evolve as we scale the number of agents, task density, and interaction topology, and where do saturation and negative collaboration appear? We summarize regularities in performance–vs–scale and communication–vs–scale curves, identify bottlenecks such as communication explosion, latency, and cascading failures, and survey structural priors (sparsity, hierarchy, modular roles) that enable migration across scales and domain shift.
- **RQ4: Evaluation Protocols Aligned with Decisions.** What metrics and benchmarks should be used to evaluate collaboration in a way that is directly decision-aligned for deployment? We systematize evaluation into three *orthogonal* dimensions: *Performance Evaluation* (capability & helpfulness: does it work?), *Alignment Evaluation* (safety & ethics: does it behave appropriately?), and *Trustworthiness Evaluation* (robustness, reliability & transparency: can we depend on and govern it in the wild?). We further highlight the practical tensions among these dimensions and summarize how existing benchmarks cover (or fail to cover) their respective blind spots.

### D. Organization

The remainder of this survey is organized as follows. Section II establishes the formal problem definition and notation, accompanied by a summary "take-away card" for quick reference. The subsequent sections systematically address our four research questions: Section III examines communication under constraints (RQ1); Section IV investigates credit assignment mechanisms (RQ2); and Section V explores scalability and policy migration (RQ3). To bridge design and deployment, Section VI proposes evaluation protocols aligned with these decision variables. Finally, Section VII synthesizes the surveyed literature through visual heatmaps, and Section VIII concludes with a discussion of open problems.

By grounding the question *Is collaboration worth it?* in deployment constraints, we synthesize scattered evidence into actionable guidance for both research and production.

### E. Comparison with existing surveys

As the field of Multi-Agent Systems matures, numerous surveys have synthesized its expanding landscape. As summarized in Table I, prior reviews have largely concentrated on specific methodological families or functional verticals. For instance, earlier works cataloged consensus protocols and control mechanisms [22], while more recent surveys have zoomed in on specific components such as communication protocols [18], [3], [23], security challenges [24], or specific emergent phenomena like saturation [25].

While these contributions provide essential taxonomies of *how* collaboration can be implemented, they often stop short of addressing the practitioner's fundamental dilemma: *whether* collaboration is viable under hard resource constraints. Existing literature rarely integrates the disparate costs of bandwidth, latency, and alignment into a unified decision framework. Our survey fills this gap by adopting a decision-oriented perspective. Instead of merely listing algorithms, we synthesize evidence on the net value of collaboration—linking communication efficiency, credit assignment fidelity, and scalability limits—to provide actionable guidance on designing systems that are not just capable, but deployable.

TABLE I
A COMPARISON OF OUR SURVEY WITH RELEVANT SURVEYS

| Year | Ref. | Contribution |
|---|---|---|
| 2022 | [22] | Overview of consensus in MAS including taxonomies, models, protocols, control mechanisms, and applications |
| 2024 | [26] | General introduction on MAS progress and challenges |
| 2025 | [18] | Focus on agents' communication protocols |
| 2025 | [27] | Focus on collaboration mechanisms of MAS. |
| 2025 | [25] | Focus on saturation and negative collaboration in large-scale MAS. |
| 2025 | [3] | Focus on agents' communication protocols and security. |
| 2025 | [23] | Focus on the principles, design, and applications of AI technologies in intelligent communication systems. |
| 2025 | [28] | Empirical analysis of LLM-based agent architectures on agentic benchmarks under controlled budgets; quantifies coordination trade-offs and learns a predictive architecture-selection model. |
| 2025 | [24] | Focus on security challenges for agents. |
| **Now** | **Ours** | **Focus on a decision-oriented framework that links communication cost–benefit, credit assignment, scalability, and evaluation protocols to answer when collaboration is worth it in real deployments.** |

## II. BACKGROUND & FOUNDATIONS

### A. Typical collaboration scenarios

*1) U2A:* User-to-agent (U2A) interaction defines the process by which human users issue high-level instructions to agents and receive feedback on execution results [29], [30], [27]. Conceptually analogous to standard conversational LLM interfaces, this mode involves a user submitting task descriptions, constraints, or preferences (e.g., "plan a three-day trip to Beijing within a fixed budget"), while the agent responds with clarifying questions, intermediate plans, or final outputs. In LLM-based multi-agent systems, the U2A channel typically delimits the coordination episode: an initial prompt activates the underlying agent collective, and the final aggregated response is returned to the user. Since U2A exchanges rely on natural language and lack the strict latency or reliability guarantees of low-level control links, existing literature predominantly characterizes this layer as *interaction* rather than communication in the classical networking sense. Nevertheless, ambiguity or misalignment at this interface remains critical, as it can profoundly dictate downstream multi-agent behavior.

*2) A2A:* Agent-to-agent (A2A) communication encompasses the exchange of information among agents to facilitate coordination, including negotiation, task decomposition, sub-task allocation, plan refinement, and result aggregation [31], [27], [32]. In conventional MAS, A2A channels transmit compact state variables, control inputs, or symbolic messages governed by explicit communication graphs and protocols, such as consensus algorithms, auctions, or contract nets. Conversely, in LLM-based multi-agent systems, A2A communication typically manifests as structured natural-language messages or tool-mediated calls between distinct agent roles (e.g., planner,

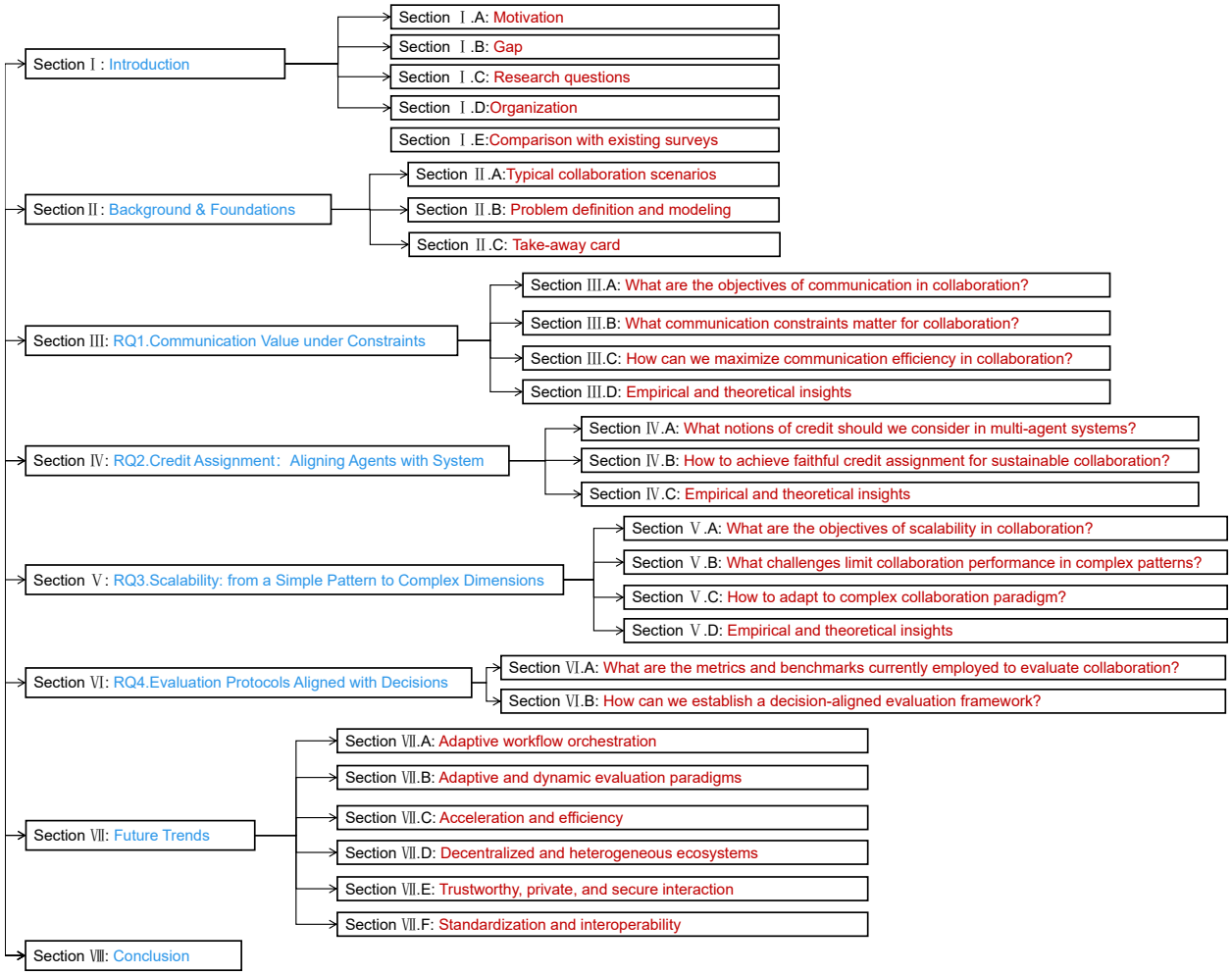


Fig. 1. Overall structure of the survey.

executor, critic, and memory manager), often orchestrated via an explicit collaboration protocol. For instance, upon receiving a travel-planning request, a coordinator agent may decompose the objective into subtasks—such as identifying attractions, verifying weather conditions, and booking accommodations—dispatch these to specialized agents, and subsequently aggregate the results. Consequently, the A2A channel serves as the primary locus of internal system coordination, where constraints on bandwidth, latency, and reliability directly dictate the richness and frequency of inter-agent interaction.

*3) A2E:* Agent-to-environment (A2E) communication characterizes the interaction between agents and external entities—such as tools, web services, simulators, and knowledge bases—facilitated by standardized protocols [33], [34], [35]. In traditional control and robotics, this interaction typically manifests through sensing and actuation channels or queries directed at shared infrastructure (e.g., maps, traffic servers, or weather APIs). Conversely, within LLM-based multi-agent systems, A2E assumes a pivotal role: agents actively invoke tools for web searches, database lookups, code execution, or third-party API calls, subsequently integrating the outputs into their decision-making processes. For example, prior to delegating subtasks, a coordinator might query a weather API for current conditions in Beijing, while downstream agents interact with booking services or retrieve tourism data from document stores. Although these A2E links lack the social dimension of inter-agent dialogue, they are essential components of the communication architecture; their availability, latency, error profiles, and access policies fundamentally constrain the information available to agents and determine the reliability of task execution.

### B. Problem definition and modeling

This subsection establishes the core concepts and notation used throughout the paper. We aim to provide a minimal, task-agnostic formalization of multi-agent systems that allows us to precisely define the notion of "collaboration" between agents.

*1) Agents:* An *agent* is an autonomous decision-making entity that repeatedly interacts with an environment. At each time step $t = 0, 1, \ldots, T$, the agent:

1) receives an *input* $x_{i,t}$ from the environment (and potentially from other agents);
2) processes this input via an internal decision mechanism;
3) produces an *output* $y_{i,t}$ that may affect the environment and/or other agents.

Formally, we model agent $i$ as a tuple

$$a_i = (\mathcal{X}_i, \mathcal{Y}_i, m_i, o_i), \tag{1}$$

where:

- $\mathcal{X}_i$ is the input space;
- $\mathcal{Y}_i$ is the output space;
- $m_i$ is the internal decision mechanism (e.g., a policy, planner, or LLM with tools), which induces a (possibly stochastic) decision rule

$$\pi_i : \mathcal{H}_i \to \Delta(\mathcal{Y}_i), \tag{2}$$

  mapping the agent's internal history $h_{i,t} \in \mathcal{H}_i$ (comprising past inputs, outputs, and internal states) to a distribution over outputs;
- $o_i$ is the agent's *objective*, specifying the desired outcomes.

At time $t$, the agent's behavior is governed by

$$y_{i,t} \sim \pi_i(h_{i,t}; m_i, o_i), \tag{3}$$

where $h_{i,t}$ updates based on $x_{i,t}$ and $y_{i,t}$ according to a specific update rule. We deliberately keep this definition implementation-agnostic: $m_i$ may be fixed or learned, and $\pi_i$ may be realized via code, neural networks, or LLM prompts.

*2) Multi-agent systems:* A *multi-agent system* (MAS) consists of a set of agents interacting with a shared environment under a defined protocol.

**Definition 1** (Multi-agent system)**.** *A multi-agent system is a tuple $\mathcal{M} = (E, A, \mathcal{I})$, where:*

- *$E$ is the environment with state space $\mathcal{S}$ and (possibly stochastic) transition dynamics;*
- *$A = \{a_1, \ldots, a_n\}$ is a finite set of agents;*
- *$\mathcal{I}$ is an interaction protocol specifying, for each time step $t$:*
  1) *how the environment state $s_t \in \mathcal{S}$ maps to agent inputs $x_{i,t}$;*
  2) *how agent outputs $y_{i,t}$ induce environment transitions and message routing.*

Concretely, a single interaction step proceeds as follows:

$$\begin{aligned} x_{1,t}, \ldots, x_{n,t} &= G_{\text{in}}(s_t, \mathcal{M}, \mathcal{I}), \\ y_{i,t} &\sim \pi_i(h_{i,t}; m_i, o_i) \quad \text{for } i = 1, \ldots, n, \\ s_{t+1} &\sim G_{\text{env}}(s_t, y_{1,t}, \ldots, y_{n,t}), \end{aligned} \tag{4}$$

where $G_{\text{in}}$ and $G_{\text{env}}$ are determined by $E$ and $\mathcal{I}$. A full *trajectory* (or episode) of length $T$ is denoted by

$$\tau = (s_0, \mathbf{x}_0, \mathbf{y}_0, s_1, \mathbf{x}_1, \mathbf{y}_1, \ldots, s_T), \tag{5}$$

where $\mathbf{x}_t = (x_{1,t}, \ldots, x_{n,t})$ and $\mathbf{y}_t = (y_{1,t}, \ldots, y_{n,t})$. This abstract setup encompasses diverse instantiations, including multi-agent reinforcement learning (MARL), distributed systems, and LLM-based agentic workflows.

*3) Objectives:* We distinguish between *individual* and *system-level* objectives.

*a) Individual objectives:* Each agent $a_i$ possesses an objective functional $o_i : \mathcal{T} \to \mathbb{R}$, mapping a trajectory $\tau$ to a scalar value $o_i(\tau)$ that the agent seeks to maximize. Often, $o_i$ takes the form of a cumulative reward:

$$o_i(\tau) = \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t r_{i,t} \,\middle|\, \tau\right], \tag{6}$$

where $r_{i,t}$ is the per-step reward and $\gamma \in [0, 1]$ is a discount factor. However, our analysis holds for general objective forms.

*b) System-level objective:* From an external design perspective, the overall performance is captured by a system-level objective $O_{\text{sys}} : \mathcal{T} \to \mathbb{R}$. Typical choices include:

- aggregated individual rewards (e.g., $O_{\text{sys}}(\tau) = \sum_i o_i(\tau)$);
- global performance metrics (e.g., success rate, latency, cost);
- multi-objective criteria combining utility with safety or fairness constraints.

The central inquiry of this paper is how interaction patterns influence $O_{\text{sys}}$.

*4) Inputs, outputs, and communication channels:* To rigorously define collaboration, we decompose agent inputs and outputs into environmental and communicational components.

For agent $i$ at time $t$, the input is decomposed as:

$$x_{i,t} = (o_{i,t}^{\text{env}},\ c_{i,t}^{\text{in}}), \tag{7}$$

where $o_{i,t}^{\text{env}}$ is the environment observation (e.g., sensor data) and $c_{i,t}^{\text{in}}$ is the *communication input*—information

explicitly transmitted by other agents or a coordination mechanism.

Similarly, the output is decomposed as:

$$y_{i,t} = (a_{i,t},\ c^{\text{out}}_{i,t}), \tag{8}$$

where $a_{i,t}$ is the *environmental action* and $c^{\text{out}}_{i,t}$ is the *communication output*.

The protocol $\mathcal{I}$ governs how $c^{\text{out}}_{i,t}$ is routed and transformed into $c^{\text{in}}_{j,t'}$ for other agents. In LLM-based systems, these components typically manifest as text strings within prompts and generated responses. This decomposition allows us to differentiate between purely environment-driven coupling (via $o^{\text{env}}$ and $a$) and explicit coordination (via $c^{\text{in}}$ and $c^{\text{out}}$).

*5) Collaboration:* We define *collaboration* operationally: agents collaborate when they intentionally leverage information about one another to inform their behavior, aiming to improve task outcomes. To formalize this, we first establish a non-collaborative reference point.

*a) Non-interacting baseline:* Consider a system where agents react solely to the environment. In this *non-interacting baseline*, we require that:

$$c^{\text{in}}_{i,t} = \varnothing \quad \text{for all } i, t, \tag{9}$$

and that each policy $\pi_i$ depends only on its own environmental history $h^{\text{env}}_{i,t} = (o^{\text{env}}_{i,0:t},\ a_{i,0:t-1})$:

$$y_{i,t} \sim \pi_i\big(h^{\text{env}}_{i,t}\big). \tag{10}$$

While agents may affect each other indirectly through the shared environment $E$, they do not reason about explicit information regarding other agents.

*b) Defining collaboration:* We state that a design "adds collaboration" when it departs from this baseline.

**Definition 2** (Collaboration)**.** *Given a multi-agent system $\mathcal{M}$ and the non-interacting baseline, agents $\{a_i\}_{i=1}^n$ are* engaged in collaboration *if at least one of the following holds:*

1) ***Explicit communication:*** *There exist $i, t$ such that $c^{in}_{i,t} \neq \varnothing$ and depends on another agent's output. Here, $\pi_i$ conditions on information intentionally provided by peers.*
2) ***Other-aware decision rules:*** *The internal mechanism $m_i$ is constructed to explicitly model, predict, or negotiate with other agents (e.g., via joint planning), exceeding the capacity of the non-interacting policy form.*
3) ***Explicitly shared objectives:*** *Objectives $\{o_i\}$ are shaped to reference other agents' outcomes (e.g., team rewards), and agents condition their decisions on this coupling rather than optimizing purely local metrics.*

*c) Cost of collaboration:* Collaboration incurs costs, including communication overhead (bandwidth, latency), coordination overhead (negotiation time), computational burden (joint reasoning), and safety risks. We aggregate these into a cost functional:

$$C_{\text{collab}} : \mathcal{T} \to \mathbb{R}_{\geq 0}. \tag{11}$$

This assigns a non-negative cost to each trajectory, representing the resources consumed by the collaborative mechanisms.

*d) The net effect of collaboration:* We view any departure from the non-interacting baseline as a *collaboration design* that induces (i) a new trajectory distribution $\tau \sim \mathcal{M}_{\text{collab}}$ and (ii) a collaboration cost $C_{\text{collab}}(\tau)$.

Our ultimate object of study is the *net effect of collaboration* on the system-level objective, defined as:

$$\begin{aligned}\Delta \ = & \underbrace{\mathbb{E}_{\tau\sim\mathcal{M}_{\text{collab}}}\big[O_{\text{sys}}(\tau) - C_{\text{collab}}(\tau)\big]}_{\text{with collaboration}} \\ & - \ \lambda\, \underbrace{\mathbb{E}_{\tau\sim\mathcal{M}_{\text{base}}}\big[O_{\text{sys}}(\tau)\big]}_{\text{non-interacting baseline}}.\end{aligned} \tag{12}$$

The central question of this paper is:

*Under what conditions, and through which collaboration designs, does the net effect $\Delta$ become positive, negligible, or negative?*

### C. Take-away card

Building upon the theoretical foundations and mathematical formulations established in the previous section, we distill the complexity of multi-agent collaboration into a unified conceptual framework. Figure 2 presents our "take-away card", illustrating the **Lifecycle of Collaborative Net Value**. This framework shifts the perspective from merely optimizing individual metrics to maximizing the holistic Net Value of the system. It posits that effective collaboration is a continuous cycle where communication costs are balanced against credit distribution, systems are stress-tested for scalability, and outcomes are rigorously evaluated to inform future iterations. This visual roadmap serves as the organizational spine for the remainder of this survey.

## III. RQ1. Communication value under Constraints

Multi-agent communication is often introduced with the promise of better coordination: agents can share local observations to overcome partial observability, exchange intent to avoid conflicts, and synchronize plans to exploit complementarities in their capabilities. In simulation, such communication has been shown to unlock substantial

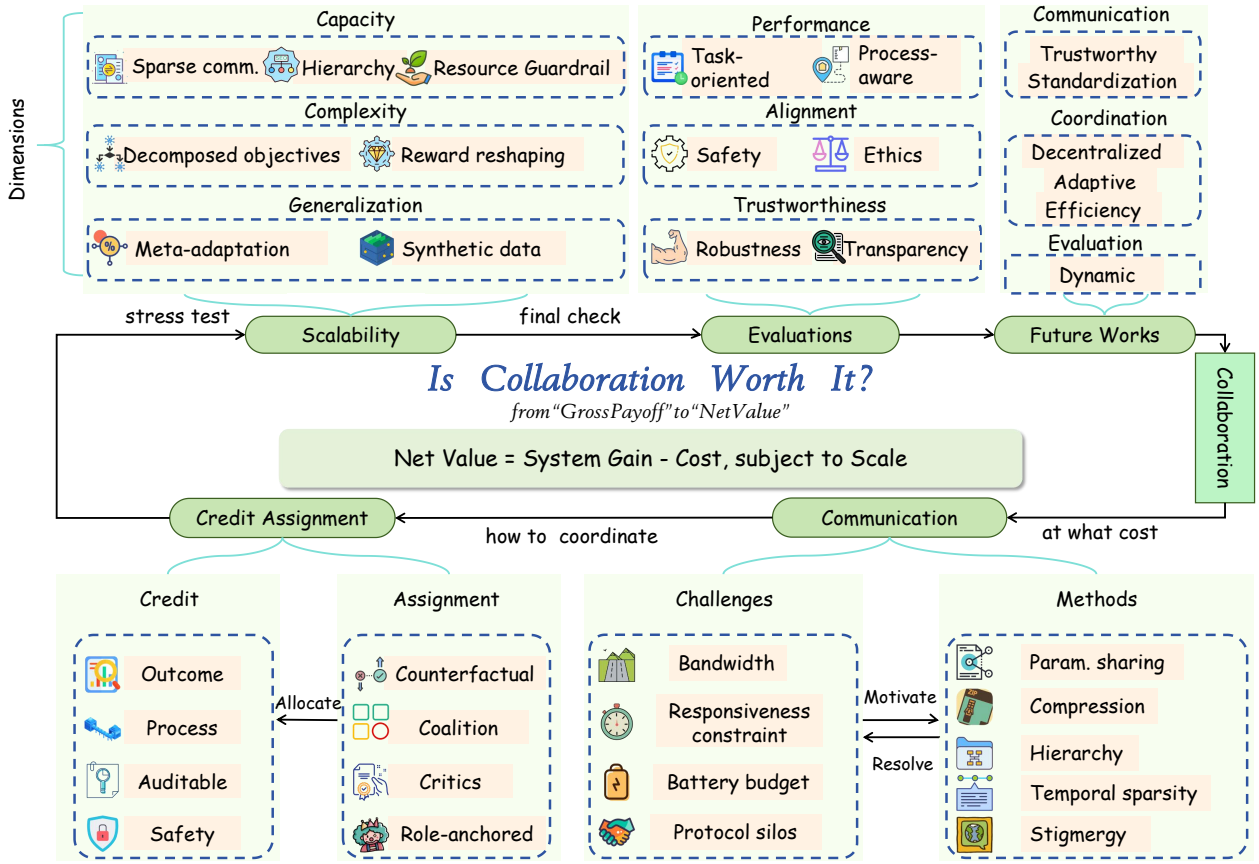


Fig. 2. **The Lifecycle of Collaborative Net Value.** This framework redefines collaboration as a pursuit of *Net Value*. In this cycle, **Communication** represents the inherent cost and constraints, while **Credit Assignment** serves as the "how-to" mechanism for distributing rewards. The system undergoes stress testing through **Scalability** and final validation via **Evaluations**. Finally, **Future Works** emerge from these evaluations to restart the cycle, driving the continuous evolution of effective collaboration.

gains in task success, sample efficiency, and robustness across domains ranging from cooperative control to tool-using LLM agents.

However, collaboration is not free. Real deployments operate under hard budgets on bandwidth, latency, energy, and compute. Many algorithmic studies implicitly assume free, reliable broadcast channels, or report only task performance while ignoring the communication required to achieve it. As a result, it is unclear when additional messages actually move the system closer to its goal, and when they merely consume scarce resources or destabilize learning.

This section therefore asks a decision-oriented question:

> **RQ1:** Under realistic constraints, when is multi-agent communication actually worth it, and what forms of communication make the best use of limited resources?

To make this question actionable, we decompose it into three sub-questions aligned with deployment decisions:

- **RQ1.1: What are the objectives of communication in collaboration?**
  Before optimizing communication, we must first categorize its functional utility. Does messaging primarily serve to share state (resolving partial observability), negotiate intent (resolving conflicts), or distribute computation (enabling modular reasoning)? Understanding these distinct roles helps clarify *why* agents need to talk in specific task environments.
- **RQ1.2: What communication constraints matter for collaboration?**
  Theoretical models often assume ideal channels, but real-world systems face physical limitations. This question examines how specific bottlenecks—such as limited bandwidth, high latency, channel noise, and privacy requirements—impact the feasibility and design of communication protocols, shifting the

focus from "what to say" to "what can be afforded".

- **RQ1.3: How can we maximize communication efficiency in collaboration?**
  Given the trade-off between coordination gains and communication costs, we explore mechanisms designed to improve the signal-to-noise ratio. This includes strategies for learning succinct representations, gating mechanisms that silence redundant messages, and topology optimization that restricts interaction to relevant neighbors only.

### *A. What are the objectives of communication in collaboration?*

Communication serves as the foundational infrastructure in Multi-Agent Systems (MAS), bridging the gap between isolated individual intelligence and cohesive collective intelligence. Based on our review of recent literature, we categorize the objectives of communication into three progressive levels: *Perception Extension*, *Intent Synchronization*, and *Knowledge Transfer*.

*a) Perception extension:* The fundamental role of communication lies in mitigating the limitations of partial observability inherent in MAS. In complex environments, individual agents are often constrained by restricted fields of view. Communication enables the sharing of local observations to construct a comprehensive global state representation. Distributed agents, such as networked radars [36], multi-camera surveillance systems [37], connected autonomous vehicles [38], [39], and swarm robots [40], [41], [42], synchronize data to achieve situational awareness over large terrains, far exceeding the capabilities of any single node.

Beyond merely expanding coverage, recent research emphasizes enhancing the robustness of collaborative perception against real-world imperfections. Addressing challenges like pose noise and motion blur in vehicle-to-everything (V2X) systems, MRCNet [39] employs multi-scale robust fusion strategies. Similarly, SCOPE [38] aggregates spatio-temporal cues to effectively compensate for localization errors during agent collaboration. Furthermore, the paradigm of perception extension is evolving from direct sensor fusion to generative state estimation in embodied AI. As exemplified by COMBO [43], generative world models are leveraged to reconstruct holistic world states from partial egocentric observations, enabling agents to mentally simulate global dynamics and plan cooperatively even when direct visual access is limited.

*b) Intent synchronization:* Beyond sharing raw data, communication is critical for synchronizing future actions. This role involves transmitting intents [44], [45], [46], and intermediate plans [47], [48] to avoid conflicts and reduce redundant efforts [49], [50]. Through explicit message passing, agents can negotiate and reach a consensus on joint actions without centralized control.

This mechanism is particularly evident in LLM based multi-agent frameworks, where synchronization often mimics human organizational workflows. For instance, in collaborative software engineering pipelines [51], [52], [53], [54], role-specific agents—acting as product managers, architects, and engineers—do not merely execute instructions in isolation; they engage in iterative dialogue to align their mental models of the system architecture before a single line of code is written [44], [45]. By sharing design documents or API specifications as explicit statements of intent, these agents preemptively resolve structural discrepancies that would otherwise lead to costly integration failures or logical bugs downstream [46].

Similarly, in embodied scenarios and complex gaming environments, intent synchronization serves as a critical collision-avoidance mechanism in the decision space. In cooperative tasks such as Overcooked or autonomous traffic negotiation, agents broadcast their immediate subgoals to peers. This explicit disclosure allows other agents to prune conflicting branches from their own planning trees, ensuring that the team avoids physical deadlocks and redundant resource usage [49], [50]. Furthermore, recent hierarchical planning approaches demonstrate that sharing high-level intermediate plans—rather than low-level primitive actions—allows agents to coordinate over longer horizons, effectively "locking in" compatible strategies while maintaining the flexibility to adapt local execution details [47], [55], [56].

*c) Knowledge transfer:* At a higher cognitive level, communication facilitates social learning, allowing agents to transcend individual trial-and-error by leveraging collective experience. Experienced agents—often acting as teachers or mentors—compress their learned value functions or decision-making processes into messages, natural language descriptions, or abstract embeddings to guide novice agents. This mechanism significantly accelerates the convergence of the entire system and enables the emergence of sophisticated cooperative behaviors that would be computationally prohibitive to discover individually.

Technically, this is predominantly realized through **Teacher-Student frameworks**, which unify distinct learning paradigms. In the context of **Knowledge Distillation** [57], [58] and **Semi-Supervised Learning** [59], [60], [61], teachers transfer "dark knowledge" via soft targets or pseudo-labels to regularize student training. For instance, classic works like Hinton et al. [62]

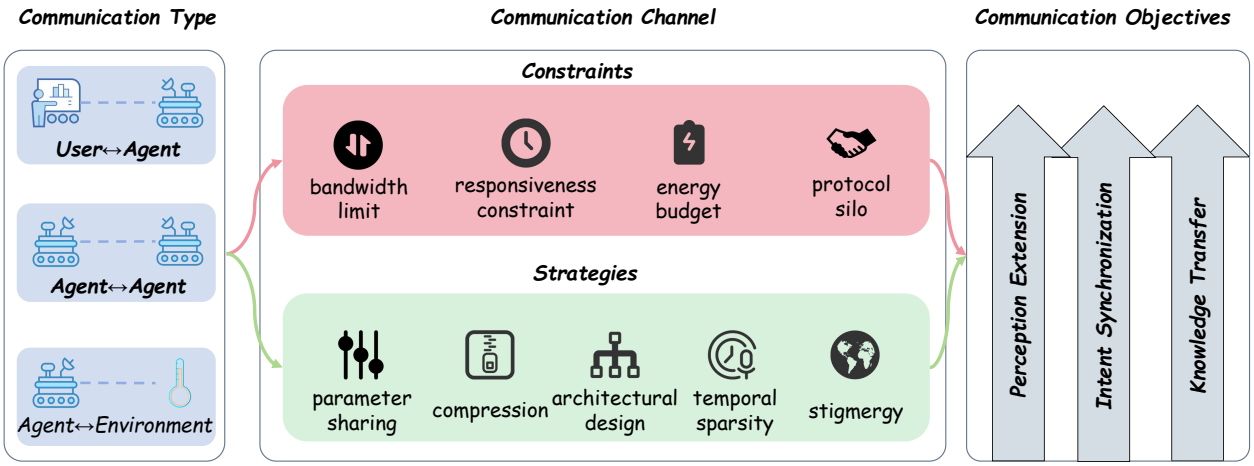


Fig. 3. A taxonomy of communication challenges, strategies, and collaborative objectives in multi-agent systems.

demonstrate how temperature-scaled logits can convey structural relationships between classes, while methods like FixMatch [63] utilize high-confidence predictions from a teacher model to guide learning on unlabeled data. Active Teacher [60] enhances semi-supervised object detection by iteratively augmenting the label set with unlabeled examples selected based on their difficulty, information, and diversity. Dual Teacher [61] addresses the weight coupling issue in semi-supervised semantic segmentation by employing two temporary teachers that alternately generate pseudo-labels to train the student model.

This paradigm extends to Domain Adaptation, where agents transfer policies across differing distributions to mitigate domain shifts [64]. In computer vision, Versatile Teacher (VT) [65] enhances cross-domain object detection by using class-specific pseudo-labels to guide instance-level alignment, overcoming the limitations of treating all classes equally. Similarly, in speech recognition, Teacher/Student learning [66] enables adaptation without target transcriptions by leveraging unlabeled parallel data—such as clean versus noisy speech—using the teacher's posterior probabilities to robustly train the student model. Specific to **Reinforcement Learning**, transfer manifests as *Action Advising* [67], [68]. For instance, Torrey and Taylor [68] propose a teacher-student reinforcement learning framework where the teacher accelerates learning by strategically budgeting a limited number of action suggestions to the student. Silva et al. [69] propose a peer-to-peer advising framework for multiagent reinforcement learning where agents learning simultaneously in a shared environment can exchange advice to accelerate learning, even without an expert teacher.

Furthermore, the advent of LLM-based agents has elevated knowledge transfer from numerical signals to semantic reasoning sharing. In these systems, agents do not just share *what* to do, but *why* to do it. For instance, in multi-agent debate or reasoning tasks, agents can articulate their chain-of-thought (CoT) processes, allowing peers to critique, refine, and adopt superior logical paths [70], [71]. This form of semantic exchange allows for the rapid propagation of "discovered skills"—such as a new coding pattern or a logical deduction strategy—across the population without the need for gradient updates. Additionally, research in emergent communication demonstrates that agents can autonomously develop specialized protocols to describe novel concepts in the environment, effectively creating a shared culture of knowledge that persists beyond the lifespan of any single agent.

### *B. What communication constraints matter for collaboration?*

In this section, we will introduce some common and important constraints in agent collaborations.

*a) Bandwidth limit:* From an information-theoretic perspective, a shared communication channel can only reliably carry a limited amount of information per unit time: by Shannon's source-coding and channel-capacity theorems [72], the average number of bits required per symbol is lower-bounded by the source entropy, and the maximum symbol rate is upper-bounded by the physical bandwidth and modulation scheme. In multi-agent systems this translates into a hard cap on how many symbols agents can collectively inject into the channel per second, given a fixed carrier bandwidth and message alphabet. When messages are continuous,

this constraint is often expressed in terms of *message entropy*: high-entropy messages require more bits to encode, whereas quantized, low-entropy messages can be compressed more aggressively under the same bandwidth budget, as made explicit in recent work that estimates the discrete entropy of quantized message streams and uses it as a proxy for bandwidth consumption in MARL [73], [74]. When communication is modeled at the level of symbols per time step, bandwidth instead appears as a limit on the *communication rate*: given a sampling frequency, message length, and number of agents, only a certain fraction of time steps can involve sending or receiving without exceeding the available bit-rate. Empirically, these bandwidth limits are particularly salient in wireless and embedded settings such as UAV swarms, underwater robots, or IoT-scale multi-agent systems. For example, multi-UAV networks often rely on low-rate long-range radios or ad hoc wireless links with tight throughput budgets [9], [75]; underwater robot teams typically communicate over acoustic modems whose usable data rates can fall well below 100 bps over practical ranges [76]; and many IoT-scale deployments where end devices are resource-limited [77], [78]. Under such conditions, agents cannot simply broadcast rich messages at every decision step, and bandwidth becomes a first-class constraint shaping whether, how often, and how verbosely they can communicate.

*b) Responsiveness constraint:* Beyond raw throughput, real communication channels also impose non-negligible end-to-end latency: messages incur propagation, queuing, medium-access, and processing delays before they influence another agent's decision. From a control-theoretic viewpoint, such delays effectively turn information into *stale* observations or actions, which can destabilize feedback loops and degrade consensus and tracking performance, especially in fast second-order or higher-order dynamics [79], [80], [81]. In discrete-time multi-agent systems, latency can be modeled as a fixed or random number of steps between message generation and reception, so that each agent reasons over delayed neighbor states or delayed control inputs rather than the current ones.

In learning-based settings, this manifests as partial observability and non-stationarity induced by outdated information, and the effective planning horizon must absorb both the environment dynamics and the communication delay. Empirically, latency constraints are prominent in wide-area or multi-hop wireless MAS, such as UAV networks relaying information over several hops or congested links [7], [82], and underwater acoustic systems where low propagation speed leads to hundreds of milliseconds or seconds of delay over practical distances [83]. Similar issues arise in emerging LLM-based multi-agent systems, where even a single high-level message already incurs substantial inference and tool-call latency. When several LLM agents interact over multiple rounds, this per-step cost is effectively multiplied across agents and dialogue turns, so a single joint decision can take orders of magnitude longer than in a comparable single-agent pipeline [84], [85], [86]. Under such conditions, agents cannot assume synchronous, near-instantaneous exchanges of language messages, and latency becomes a first-class constraint on how many communication rounds are affordable within a given real-time budget.

*c) Battery budget:* Real communication is constrained by *energy*: transmitting, receiving, and processing messages all consume power, which is often scarce or tightly budgeted. From a communication-theoretic standpoint, reliable data transfer over a noisy channel requires a minimum energy per bit, and operating at low signal-to-noise ratios or over long distances quickly amplifies the power required to sustain a given rate and reliability [72]. In embedded and battery-powered platforms, radios, modems, and on-board computation typically dominate the energy footprint, so every additional packet or bit of information has a tangible cost in terms of battery life and mission duration [87]. As a result, communication cannot be treated as "free": agents must reason not only about *what* to say and *when*, but also about *whether* the informational benefit of a message justifies its energetic cost.

In networked control and consensus, these energetic constraints are often abstracted through *event-triggered* or *self-triggered* communication schemes, where agents only transmit when state deviations exceed certain thresholds, rather than at every time step [88], [89]. Such designs explicitly trade off convergence rate and performance against reduced communication energy, and stability analyses impose conditions under which sparse transmissions still guarantee consensus or formation-keeping. In wireless sensor networks and IoT-style MAS, energy-aware MAC protocols, clustering, and data aggregation methods are used to minimize radio usage; agents may compress, quantize, or fuse local observations before transmission, or rotate communication roles to balance battery drain across the network [87], [90]. Here, the energy budget directly limits how frequently and how richly agents can exchange information, forcing coordination strategies that are robust to infrequent, low-rate updates.

Learning-based multi-agent systems inherit the same constraints, but now energy costs also shape the *training* and *adaptation* process. In MARL for wireless and mobile networks, agents must learn communication and

control policies under joint constraints on throughput, delay, and power; optimal behaviors often involve turning radios off when idle, using low-power modes, or dynamically adjusting transmission power based on channel conditions [15], [91]. Communication-learning methods therefore incorporate explicit costs or penalties on message length, frequency, or power usage, encouraging agents to discover compact protocols, sparse signaling patterns, or implicit coordination through actions rather than chatty explicit messages [92], [93]. Empirically, energy-aware designs are critical in MAS such as UAV swarms [7],underwater and subterranean robots [83], where acoustic or low-frequency communication is extremely energy-intensive and thus used sparingly [83], [94] and large-scale sensor networks [95]. Emerging LLM-based multi-agent systems face analogous, if higher-level, energy and cost constraints. Each message may trigger expensive model inferences, tool calls, or remote API invocations, consuming significant computational power and monetary budget in cloud environments [96].

*d) Protocol Silos:* Even when agents are theoretically capable of collaboration, their interaction is strictly bounded by the *interoperability protocols* that govern the exchange of $c^{\text{out}}$ and $c^{\text{in}}$. As recent surveys indicate, the current agent landscape is fragmented into distinct "protocol silos" that impose structural barriers to collaboration [31], [3]. These silos manifest in three critical dimensions that constrain the system-level objective $O_{\text{sys}}$:

- **Architectural Mismatch:** Collaboration is often blocked by incompatible topological assumptions. For instance, agents built on the *Model Context Protocol (MCP)* operate within a strict client-host relationship designed for local context injection [31], [97], whereas agents using the *Agent Network Protocol (ANP)* assume a decentralized, peer-to-peer topology based on Distributed Identifiers (DIDs) [98]. An agent in the MCP silo effectively cannot "see" or collaborate with an agent in the ANP silo without complex bridging middleware, partitioning the collaboration graph.
- **Discovery and Binding Friction:** Before information can be shared, agents must locate one another. Protocol silos enforce disjoint discovery mechanisms: *Agent-to-Agent (A2A)* protocols rely on capability-based "Agent Cards" and capability negotiation [99], [100], while *Agent Communication Protocol (ACP)* relies on registry-based lookups or offline manifests [31], [101]. These distinct discovery substrates mean that the cost of collaboration $C_{\text{collab}}$ includes not just message transmission, but the significant overhead of cross-protocol translation and adapter maintenance.
- **Rigid Interaction Lifecycles:** Finally, protocols impose mandatory state machines—spanning Creation, Operation, and Termination—that dictate the granularity of collaboration. Security requirements, such as the specific handshake and authentication phases required by A2A or the JSON-RPC schema enforcement in MCP [31], [102], introduce mandatory latency and round complexity. These "lifecycle costs" act as a tax on collaboration, rendering fine-grained, high-frequency coordination prohibitively expensive in heterogeneous environments.

Consequently, the "Protocol silos" effectively restrict the policy space $\Pi$: agents are often forced to optimize local objectives $o_i$ simply because the protocols required to align with global objectives are inaccessible or too costly to bridge.

### *C. How can we maximize communication efficiency in collaboration?*

*a) Parameter sharing:* A natural way to reduce communication costs while preserving the benefits of collaboration is to *move part of the collaboration into shared representations*, so that agents can implicitly coordinate through common parameters or latent spaces rather than through lengthy explicit messages. In multi-agent deep RL, classical parameter sharing allows agents to reuse a single policy or value network, substantially reducing the number of trainable parameters and enabling algorithms to scale to many agents with shorter training times and improved sample efficiency [103]. However, naively forcing all agents to share exactly the same network can degrade performance when their roles, abilities, or goals differ.

Recent work therefore focuses on *structured* parameter sharing. Christianos et al. [103] propose to automatically partition agents into groups based on their abilities and goals, such that only appropriate subsets share parameters. This combines the efficiency of sharing with the expressiveness of multiple specialized networks. Terry et al. [104] systematically formalize "agent indication", that is, agent-specific identifiers appended to observations, to enable a single shared network to implement distinct per-agent policies and to extend parameter sharing to heterogeneous observation and action spaces. They provide theoretical guarantees of convergence to optimal policies together with supporting empirical results.

Beyond standard RL, shared internal representations can also reside in continuous latent spaces rather than in parameter vectors or text. LatentMAS [105] demonstrates that LLM-based multi-agent systems can collaborate via a shared latent working memory that stores last-layer hidden states from each agent. Agents then read

from and write to this common latent space instead of exchanging natural language, which preserves rich internal information and yields substantial reductions in output tokens and inference time, while also improving accuracy. Related ideas arise in hierarchical and task-level structures. Feudal latent-space exploration [106] introduces a global latent structure, learned by a commander, to coordinate agent exploration in a low-dimensional shared space, improving both efficiency and coordination quality relative to independent exploration. DSRA [107] learns shared latent representations of subtasks together with agent-specific ability encoders, allowing agents to specialize in different subtasks while still sharing knowledge and experience among agents assigned to similar roles. On the computational front, KVCOMM [108] leverages shared internal states to accelerate inference by enabling cross-context KV-cache communication. It identifies and compensates for the representation offsets caused by different agent prefixes, allowing agents to reuse cached internal states for overlapping contexts and reducing prefilling latency by up to $7.8\times$.

Across these lines of work, the common theme is to compress and centralize coordination into shared parameters or latent representations, thereby reducing explicit communication overhead without sacrificing, and in some cases improving, system-level collaborative performance.

*b) Compression:* A complementary way to reduce communication cost, beyond parameter or representation sharing, is to directly *compress the interaction protocol* between LLM agents—either by sparsifying who talks to whom, structuring how information is aggregated, or learning more concise communication patterns. In multi-agent debate, Li et al. [109] systematically study the effect of communication connectivity and show that a brute-force fully connected topology is often unnecessary: carefully designed *sparse* debate graphs, where each agent only exchanges messages with a limited subset of peers, can match or even outperform fully connected debates while substantially lowering computational and token costs. Liu et al. [110] further reduce debate overhead by introducing GroupDebate, which partitions agents into multiple debate groups that first reason internally and then share *intermediate summarized results* across groups. This grouping scheme both prunes the number of pairwise interactions and compresses cross-group communication into shorter summaries, yielding up to 51.7% reduction in total tokens while potentially improving accuracy by as much as 25%. Going beyond hand-designed topologies and grouping, OPTIMA treats the communication protocol itself as an object of optimization: it uses an iterative generate–rank–select–train paradigm with a reward that explicitly balances task performance, token efficiency, and readability, and trains LLM-based agents (via SFT, DPO, and hybrids) to adopt interaction patterns that convey the same task-relevant information with far fewer tokens [111]. On multi-agent tasks requiring heavy information exchange, OPTIMA achieves up to $2.8\times$ performance gains using less than 10% of the tokens of vanilla LLM-based MAS, illustrating that *learned protocol compression* can simultaneously improve collaborative effectiveness and communication efficiency.

*c) Architectural design:* Beyond compressing individual messages, a substantial body of work reduces communication cost and enhances interoperability by *architecting the multi-agent system itself*. This architectural optimization operates along two distinct dimensions: **structural topology**—designing who talks to whom to minimize redundancy—and **protocol governance**—structuring the communication stack to resolve compatibility silos.

Regarding structural topology, the core idea is to design the interaction graph so that useful coordination emerges with far fewer interactions. This idea is classical in control-theoretic MAS: for continuous-time average consensus, Rafiee et al. formulate the optimal design of communication graphs as a mixed-integer semidefinite program over strongly balanced digraphs, either (i) maximizing convergence speed given a hard budget on the number of links, or (ii) minimizing the number of links subject to a required convergence rate in the presence of delays [112]. Related architectural considerations arise in distributed energy management, where Khan et al. replace a monolithic centralized controller with a multi-agent, game-theoretic distributed architecture for coordinating distributed generators and loads [113]. Their simulations under renewable fluctuations, seasonal demand changes, and grid disturbances show that an appropriately decentralized structure can improve robustness and control performance while avoiding the communication and fault-tolerance bottlenecks of a single central coordinator.

These structural principles are increasingly mirrored in LLM-based MAS. AutoHMA-LLM [85] proposes a multi-tier heterogeneous architecture for drone–robot–device systems, combining a cloud LLM as a global planner with device-specific LLMs and generative agents as local executors. By pushing high-level planning and scheduling to the cloud layer and leaving low-level control and feedback to edge agents, the framework reduces redundant cross-device coordination: in logistics, inspection, and search-and-rescue scenarios it achieves a 46% reduction in communication steps and a 31% decrease in token usage and API calls, while improving task completion accuracy by 5.7%. Other work focuses more directly on

TABLE II
COMMUNICATION CONSTRAINTS VS. EFFICIENCY METHODS IN RQ1.3. COLUMNS DENOTE RESOURCE CONSTRAINTS (BW: BANDWIDTH, RESP.: RESPONSIVENESS, BATT.: BATTERY, PROTO.: PROTOCOL SILOS). RATINGS: ++ STRONG BENEFIT, + BENEFIT, 0 NEUTRAL, ± CONTEXT-DEPENDENT, – HARMFUL.

| Method | Core idea | BW | Resp. | Batt. | Proto. | Pitfall |
|---|---|---|---|---|---|---|
| Parameter sharing | Share policy parameters or latent representations | + | ± | + | – | Synchronization overhead; reduces agent diversity and robustness |
| Compression | Compress interaction protocols directly | ++ | ± | ± | ± | Encoding/decoding compute cost; information loss or error accumulation |
| Architectural design | Learn task-adaptive communication topology | + | + | + | + | Single point of failure; bottleneck nodes; high design complexity |
| Temporal sparsity | Trigger interactions based on confidence/complexity | ++ | ± | ++ | 0 | Stale information harms fast dynamics; hard to tune triggering thresholds |
| Stigmergy | Coordinate indirectly via shared environment | + | + | + | ++ | Requires shared infrastructure; consistency, garbage collection, and security risks |

*topology design* in LLM agent societies. G-Designer [114] models the MAS as a graph and uses a variational graph auto-encoder, conditioned on a task-specific virtual node, to generate task-aware communication topologies: it learns which agents should be connected and how densely, yielding pipelines that maintain or improve performance (e.g., on MMLU and HumanEval) while cutting unnecessary token-heavy interactions. Complementary pruning-based approaches start from rich or over-connected pipelines and *remove* redundant structure. AgentDropout [115] optimizes the adjacency matrices of communication graphs across rounds to identify and drop redundant agents and edges, leading to an average 21.6% reduction in prompt tokens and 18.4% reduction in completion tokens alongside performance gains. AgentPrune [116] formalizes communication redundancy in spatial–temporal message-passing graphs and performs one-shot pruning of superfluous or even malicious messages, producing economical topologies that match state-of-the-art performance at a fraction of the monetary cost (e.g., \$5.6 vs. \$43.7) and achieve 28.1%–72.8% token reduction while improving robustness to adversarial agents. Moving further toward full automation, ARG-Designer [117] reframes MAS construction itself as conditional autoregressive graph generation: given a natural-language task description, it sequentially decides how many agents to instantiate, which roles to assign, and which communication links to create. This generative approach jointly designs system composition and topology, yielding task-tailored collaboration structures that attain state-of-the-art performance with significantly higher token efficiency and better extensibility. Across these lines of work, the common theme is that *who is connected to whom, at what level of abstraction, and with what sparsity* is a first-class design lever: by optimizing architectures and topologies, MAS can substantially reduce communication overhead—sometimes by half or more—while preserving or even improving collaborative effectiveness.

While optimizing topology addresses the efficiency of interactions, a parallel architectural challenge lies in the governance of interaction standards, ensuring *quality* and *compatibility* to prevent system fragmentation. This leads to the second dimension: addressing *protocol silos* through hierarchical governance. As LLM-based agents proliferate, ad-hoc interfaces often trap agents in isolated ecosystems, hindering cross-platform collaboration. To mitigate this, recent frameworks advocate for a *layered protocol architecture* akin to the OSI model in networking [31]. For instance, the Model Context Protocol (MCP) standardizes the connection between agents and data sources, effectively decoupling the "tool use" layer from the "reasoning" layer. Similarly, the Agent Control Protocol (ACP) focuses on the orchestration layer, defining how agents hand off tasks regardless of their underlying LLM backbones. By adopting this tiered governance—separating the transport of messages from the semantics of tool execution—architects can break down silos. This ensures that a topologically optimized agent (as discussed above) can not only communicate efficiently but also interoperate universally across heterogeneous environments.

*d) Temporal sparsity:* In addition to sparsifying *who* is connected to whom, an orthogonal line of work focuses on *when* agents should communicate, aiming for temporal sparsity so that interaction is triggered only when it is actually useful. In LLM-based multi-agent debates, Li et al. already show that reducing the number

of active communication edges within and across rounds can significantly cut computation while preserving or even improving reasoning quality [109]. However, most debate-style pipelines still run a fixed number of rounds for every query, irrespective of difficulty, which wastes computation on easy problems and may even amplify errors through unnecessary back-and-forth.

To tackle this, Eo et al. propose *Debate Only When Necessary* (DOWN), an adaptive framework that selectively activates multi-agent debate based on the confidence of an initial single-agent response [118]. If the model is sufficiently confident, the system simply returns the one-shot answer without spawning any additional agents or debate rounds; only low-confidence queries trigger multi-agent interaction, during which agents refine their outputs by referencing peers' responses and associated confidence scores. This simple temporal gating of debate yields up to $6\times$ efficiency gains while matching or surpassing the accuracy of always-on debate, and importantly mitigates error propagation that can arise from debating "for its own sake" on already-correct answers. Taking a more structural approach to on-demand interaction, AgentNet [121] eliminates static communication workflows by equipping each agent with a decentralized decision-making mechanism. Instead of following a pre-defined interaction graph, each agent in AgentNet utilizes a specific Router module to evaluate incoming tasks against its own retrieval-augmented memory and expertise. For every task step, the agent autonomously decides between three actions: *Execute* (solving the task locally without communication), *Forward* (passing the task to a more suitable agent), or *Split* (decomposing the task and distributing subtasks). This mechanism ensures that communication is strictly event-triggered—occurring only when an agent explicitly determines that its local capabilities are insufficient or that a specialist is required. Consequently, the system naturally prunes unnecessary message passing, dynamically evolving the collaboration topology (a Directed Acyclic Graph) to match the immediate complexity of the problem.

Across these works, a consistent principle emerges: communication in multi-agent systems should be treated as a scarce resource. By conditioning interactions on confidence levels (as in DOWN) or capability gaps (as in AgentNet), temporally sparse protocols can drastically reduce communication overhead while retaining the benefits of collective intelligence for complex queries.

More recent work generalizes temporal sparsification to richer, structured collaborations. HyperAgent [119] models agent societies as hypergraphs and uses a variational framework with sparsity regularization to *adapt both the strength and frequency of group interactions to task complexity*. For simple tasks, the learned hypergraph becomes temporally and structurally sparse—few hyperedges are activated, and information can be aggregated in a single step within small collaboration groups—thereby avoiding unnecessary recurrent exchanges; for harder tasks, additional hyperedges and interaction steps are selectively activated to sustain deeper coordination. In distributed control and robotics, Quan et al. advocate a related "normally dormant, intervention-on-demand" moderator mechanism for adversarial environments [122]. Rather than maintaining a constantly active centralized controller, their moderator remains silent under routine conditions and only intervenes when conflict or deadlock is detected, issuing brief, high-level directives (e.g., a short coordination command in StarCraft) that unblock local policies. This event-triggered intervention achieves a favorable optimal efficiency ratio by harmonizing autonomous local decisions with rare but well-timed global coordination, illustrating how *just-in-time* communication can outperform both fully centralized and fully decentralized baselines.

Across these works, a consistent principle emerges: communication in multi-agent systems should be treated as a scarce resource in the *time domain*. By turning interactions into events that are *conditioned on confidence, state change, conflict, or task complexity*, temporally sparse protocols can reduce the number of debate rounds, messages, or coordination interventions—often by factors of two to six—while retaining, or even enhancing, overall task performance.

*e) Stigmergy:* A complementary strategy to explicit message passing is to leverage the *environment itself*—whether a physical world, an embodied simulator, or an abstract game—as the primary medium through which agents coordinate. Instead of transmitting dense tokens at every step, agents act, observe, and infer others' intentions via changes in the shared environment, often using language only at key planning or summarization points. Zhang et al. [49] build Cooperative Embodied Language Agents (CoELA) for challenging embodied multi-agent tasks with decentralized control, raw sensory observations, costly communication, and multi-objective goals. Rather than assuming a free broadcast channel or a centralized controller with global state, they embed LLMs into a cognitive-style architecture that integrates perception, memory, and action. LLM-powered agents use their commonsense knowledge and reasoning to plan locally and communicate *only when needed* in natural language, exploiting the rich state of embodied environments (CWAH, TDW-MAT) as a shared context. Experiments with GPT-4-driven CoELA show that such agents can outperform strong planning baselines while ex-

TABLE III
COMPARISON OF MULTI-AGENT COMMUNICATION STRATEGIES (CORE IDEA COLUMN REMOVED).

| Strategy | Example | Comm. Overhead | Performance | Scalability | Robustness | Impl. Complexity |
|---|---|---|---|---|---|---|
| Parameter sharing | LatentMAS [105] | **Low**: Shared latent working memory | **High**: Rich, lossless information exchange | **High**: Decoupled from agent count | **Medium**: Consistent LLM representations but vulnerable to noise | **High**: KV cache modifications and vLLM hacks |
| | DSRA [107] | **Low**: Independent action after assignment | **High**: SOTA on complex benchmarks | **High**: Sub-task specific policy sharing | **Medium**: VAE & encoder dependency | **Medium**: VAE & encoder training |
| Compression | Group-Debate [110] | **Low**: Group-wise summarization | **High**: Grouped hierarchical debate | **High**: owing to tunable hierarchical grouping | **Medium**: stable locally but sensitive to inter noise | **Low**: Simple grouping & summarizer |
| | OPTIMA [111] | **Low**: Lightweight learned protocols | **High**: Up to 2.8× performance gain | **High**: Cross-task generalization | **Medium**: Dependent on preference data quality | **High**: Iterative training (SFT+DPO) |
| Architecture design | AutoHMA-LLM [85] | **Low**: Reduced cross-device coordination | **High**: +5.7% task accuracy | **Medium**: Fixed hierarchical structure | **High**: Edge autonomy mechanism | **Medium**: Cloud-edge architecture |
| | ARG-Designer [117] | **Low**: Economical topology generation | **High**: SOTA performance | **High**: Dynamic architecture generation | **High**: Malicious message pruning | **High**: Topology model training |
| Temporal sparsity | DOWN [118] | **Low**: No interaction on easy queries | **High**: Error propagation avoidance | **High**: Independent of agent count | **Medium**: Confidence score dependency | **Low**: Confidence extractor only |
| | HyperAgent [119] | **Low**: Sparse hypergraph usage | **High**: Excellent task performance | **High**: Linear scaling with agents | **Medium**: VAE hypergraph construction | **High**: Variational hypergraph autoencoder |
| Stigmergy | CoELA [49] | **Low**: Implicit environmental coordination | **High**: Superior planning in embodied tasks | **High**: Scales with environment complexity | **High**: Robust to message loss | **Medium**: Simulator integration |
| | LLM-DTParam [120] | **Low**: Implicit simulation coordination | **High**: Automated parametrization | **Medium**: Simulation-bound modularity | **Medium**: Textual state representation | **High**: Sequential agent loop |

hibiting *emergent, efficient* communication patterns; fine-tuned open models (CoLLAMA) further demonstrate that much of the coordination burden can be offloaded onto environment-aware behavior primitives, with language used primarily for high-level negotiation and clarification. A user study reveals that natural-language interaction layered on top of environment-mediated coordination increases human trust and cooperative effectiveness, suggesting that explicit messages can be sparse yet semantically rich when grounded in a shared world.

Environment-mediated coordination is also central in abstract game-theoretic settings. Agashe et al. [123] introduce the LLM-Coordination Benchmark to study LLMs in *pure coordination* games, where payoffs depend entirely on aligning actions rather than exchanging explicit messages. In the Agentic Coordination track, LLMs act as agents that must choose actions based on *environmental cues* and payoff structures; in CoordQA, models answer questions that probe environment comprehension, theory-of-mind (ToM) reasoning, and joint planning. Results show that LLM agents perform well when coordination can be achieved by reading and reasoning over environmental variables alone, but struggle when success hinges on representing and anticipating partners' hidden beliefs and intentions. Zero-shot coordination experiments further reveal that, unlike many RL policies that overfit to specific partners or training curricula, LLM agents can often coordinate robustly with *unseen* partners purely through shared interpretation of the game description. This highlights both the promise and the current limitations of using the environment plus implicit conventions as a low-bandwidth coordination channel.

A similar pattern emerges in digital-twin and simulation-based systems. Xia et al. [120] design a multi-agent framework in which specialized LLM agents observe, reason about, and adjust parameters of a digital-twin simulator to achieve user-specified objectives. Here, the simulation environment functions as a high-fidelity, continuous feedback channel: agents propose parameter changes, observe the resulting system behavior, and iteratively refine their decisions, using language mostly for *meta-level* tasks such as hypothesis generation, explanation, and summarization. By tightly coupling language agents with an executable environment, the framework enables autonomous exploration of feasible parametrizations while keeping explicit communication overhead low—coordination largely flows through shared trajectories and state evolutions in the simulator. Across these domains, the core idea is to *push as much coordi-*

*nation as possible into the shared environment*: agents read from and write to the world (or its simulation) and reserve direct language communication for sparse, high-level exchanges, thereby improving scalability and efficiency in settings where bandwidth or token budgets are constrained.

### D. Empirical and theoretical insights

*a) Empirical studies:* Across MARL communication benchmarks and emerging LLM-based multi-agent systems, empirical evidence consistently reveals a saturation effect: performance typically improves rapidly with the introduction of minimal communication, but exhibits diminishing marginal returns as message frequency, length, or topology density increases. Beyond a task-dependent saturation point, additional communication often yields negligible gains or even induces "negative collaboration" due to signal-to-noise degradation and increased reasoning overhead. Furthermore, a distinct *communication efficiency frontier* emerges: methods that leverage structural priors—such as temporal sparsity (event-triggered communication), topological pruning (sparse/dynamic graphs), and implicit coordination (stigmergy, shared representations)—consistently dominate naive full-broadcast or dense-debate baselines, achieving comparable task success with significantly lower bandwidth, latency, and energy consumption.

*b) Decision rules:* These findings support a decision-oriented framework for designing collaborative interactions.

- *Assess the marginal value of bits:* Communication should be treated as a budget allocation problem; if the estimated marginal performance gain per additional token or message bit falls below the deployment cost (in latency or energy), practitioners should cap interaction and rely on local policy execution.
- *Match mechanism to binding constraints:* Under strict *bandwidth or token* limits, prioritize implicit coordination mechanisms (e.g., stigmergy, parameter sharing) and protocol compression over explicit messaging. Under strict *latency* limits, avoid multi-round consensus or debate loops; instead, favor hierarchical architectures or event-triggered protocols that localize decision-making.
- *Prune by default:* To ensure scalability, avoid defaulting to fully connected interaction graphs; start with sparse, role-based, or attention-pruned topologies to prevent communication explosion as the agent population grows.

## IV. RQ2. Credit Assignment: Aligning Agents with System.

LLM-based multi-agent systems often achieve system-level goals through distributed decisions across multiple agents, dialogue turns, and tool calls. This raises a key alignment challenge: outcomes are observed globally, but improvements, incentives, and accountability must be applied locally. Credit assignment addresses this by attributing system-level gains and failures to specific agents and intermediate actions, enabling targeted training, efficient routing, and precise debugging.

In these systems, "credit" is not a single quantity but a family of contribution notions grounded in observable feedback signals. Depending on the chosen valuation signal, credit can be attributed at different granularities—from high-level roles to individual messages, tool calls, or artifacts like code patches. To systematically address this alignment problem, we decompose the challenge into two sub-questions:

- **RQ2.1: What notions of credit should we consider in multi-agent systems?**
  Before assigning credit, we must define what constitutes a valuable contribution. This question explores how abstract concepts of utility are operationalized into concrete signals. It examines the spectrum of credit definitions, ranging from extrinsic outcome-based metrics (e.g., success rates, user satisfaction) to intrinsic process-based measures (e.g., information gain, logical coherence), and determines the appropriate unit of analysis—whether it be the agent, the role, or the specific artifact produced.
- **RQ2.2: How can we achieve faithful credit assignment for sustainable collaboration?**
  Once credit is defined, the challenge shifts to the mechanisms of attribution. This question investigates algorithmic and heuristic methods for distributing global rewards across local actions. It focuses on ensuring *faithfulness*—guaranteeing that agents are rewarded for genuine contributions rather than exploiting spurious correlations—and *sustainability*, ensuring that incentive structures promote long-term cooperation rather than short-sighted competition or free-riding.

### A. What notions of credit should we consider in multi-agent systems?

In LLM-based multi-agent systems, *credit* denotes an attributed notion of *contribution*: how much a component of a team interaction (an agent, a turn, or a tool action) advances the system toward its objectives. Because agent teams can reach the same final outcome through different

internal trajectories, "contribution" must be defined at the semantic level rather than assumed to be implied by fluent intermediate text. Throughout this survey, we use *credit* as a generic container for the value assigned to intermediate actions with respect to the system's goals, which may be scalar (a single notion of usefulness) or vector-valued (multiple, potentially competing objectives).

*a) Outcome contribution: Outcome contribution* defines credit in terms of **improvements in the final task result**—e.g., correctness, completeness, and user utility of the final output. It is the most intuitive credit notion because it directly matches end-user success. *However,* in LLM agent teams outcome-only credit is often **sparse and entangled**: many different internal trajectories can lead to the same final answer, making it hard to isolate which intermediate actions were actually decisive. Concrete systems nevertheless rely heavily on outcome-oriented credit to decide which behaviors to reinforce. In tool-using coding settings, outcome is often instantiated as *execution success* [12], [124], [125], so credit is assigned to the steps (or sub-agents) whose edits and tool calls lead the program into a passing state. Similarly, collaborative software-engineering frameworks such as MetaGPT [44] and AutoGen-style pipelines [45] evaluate role-specialized agents by whether the produced artifacts satisfy task requirements, making final deliverable quality the dominant notion of contribution. Debate- and jury-style systems likewise adopt an outcome view, treating an agent as contributing when its arguments *shift the final verdict* toward a more accurate decision. Empirically, Smit et al. [126] quantify the utility of multi-agent debate (MAD) primarily via end-to-end accuracy, and use auxiliary debate statistics (e.g., consensus and answer changes) to contextualize how verdict shifts arise under different protocol settings.

*b) Process contribution: Process contribution* defines credit in terms of **improvements in the interaction trajectory**—i.e., how efficiently, reliably, and coherently the team *arrives* at an answer, rather than only whether the final answer is correct. Typical process-level notions include reduced cost/latency, increased stability, improved coordination, and better deliberation quality. This notion is especially salient in LLM agent teams because system performance is often constrained by budgets and context windows, and because different trajectories can yield the same outcome with very different resource footprints and failure risks. Concrete systems instantiate process contribution in terms of *resource efficiency and interaction quality* rather than only final correctness, tracking signals such as token usage, latency, number of rounds, and communication redundancy. AgentTaxo [127] characterizes duplicated tokens as a *communication tax* (up to 86% duplication), linking process cost to repeated role instructions and recycled reasoning in verification. AgentDropout [115] optimizes the multi-agent communication graph by dropping low-utility agents/edges across rounds, reducing token consumption while preserving or improving performance. Beyond graph-level pruning, CodeAgents [128] shows that codifying multi-agent interaction into structured pseudocode can compress trajectories (lower input/output tokens) and improve planning success. In embodied collaboration, ReAd [129] replaces expensive trial-and-error feedback with advantage-based refinement, reducing environment interaction steps and LLM query rounds as explicit process gains.

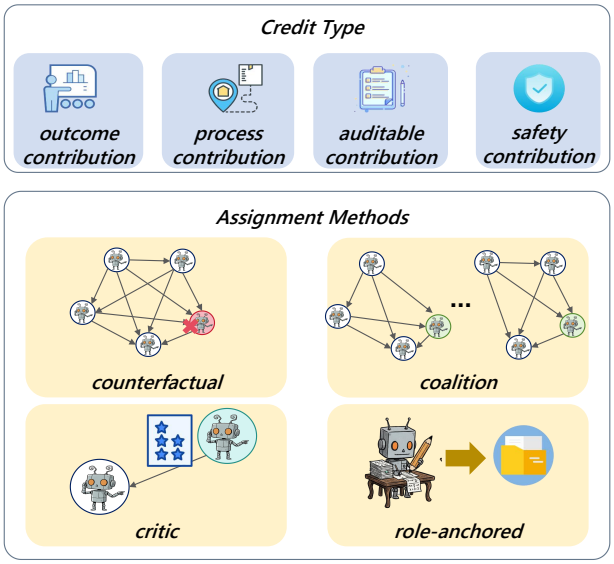


Fig. 4. Credit types and assignment methods in multi-agent systems.

*c) Auditable contribution: Auditable contribution* defines credit in terms of **improving traceability and verifiability of the team trajectory**—i.e., whether intermediate artifacts make it possible to *inspect, justify, and reproduce* how a final decision was reached. Unlike outcome contribution (which rewards being correct) or process contribution (which rewards being efficient), auditable contribution rewards producing *checkable intermediate objects*, such as structured logs, explicit rationales linked to claims, summaries that preserve key evidence, or diagnostic feedback signals that explain why replanning occurred. This notion is particularly relevant in high-stakes or regulated settings where stakeholders require post-hoc accountability, and where multi-agent interaction can otherwise devolve into opaque message passing with limited external scrutiny. Concrete systems operationalize auditability by explicitly recording and structuring

intermediate reasoning traces that can be reviewed independently of the final answer. For example, TruEDebate (TED) [130] simulates a formal pro–con debate and produces debate logs together with a synthesized debate report, which serve as auditable artifacts supporting fake-news judgments and enabling human inspection of the arguments leading to a conclusion. Similarly, ReAd [129] makes replanning decisions auditable by attaching critic-derived advantage scores and threshold-based refinement triggers to proposed actions, providing an explicit, inspectable justification for when and why a plan is revised before execution.

*d) Safety contribution: Safety/compliance contribution* defines credit in terms of **reducing harms and constraint violations** throughout the interaction. Concrete systems operationalize this notion via *pipeline-wide safeguards with measurable violation reductions*, attributing safety gains or failures to specific stages such as planning, tool use, memory operations, and final response filtering. PSG-Agent [131] implements personalized, cross-turn guardrails by generating user-specific safety criteria from interaction history and current queries, then enforcing them with specialized monitors spanning plan, tool, memory, and response stages. SafeScientist [132] similarly treats safety as an end-to-end objective in scientific agent workflows, combining prompt, collaboration, and tool-use monitoring with a final ethical reviewer, and evaluates safety improvements on SciSafetyBench. The Trust-Vulnerability Paradox study [133] formalizes safety credit as adherence to the *minimum-necessary-information* principle under varying inter-agent trust, introducing unified leakage metrics (Over-Exposure Rate and Authorization Drift) and showing that lightweight defenses can reduce over-disclosure. Finally, safety-critical multi-agent MCTS for unsignalized intersections [134] grounds safety credit in constraint satisfaction and risk-metric reductions (e.g., lower PET violations) through risk-aware pruning during search, illustrating an explicit linkage between safety objectives and intermediate decision steps.

### *B. How to achieve faithful credit assignment for sustainable collaboration?*

Given that decentralized learners update from local signals, sustainable collaboration requires that each agent's learning update is (as far as possible) *faithful* to its causal contribution to the system-level objective $O_{\text{sys}}$. Here, "fair" credit assignment refers to *perfect/accurate* attribution—i.e., the extent to which credit signals (rewards, advantages, value targets, or gradients) track an agent's marginal impact on $O_{\text{sys}}$—rather than distributive or normative fairness. Building on the three challenge dimensions in RQ2.2, we organize methods and desiderata for improving credit assignment into three corresponding layers: statistical (estimators), behavioral (strategic incentives and stability), and architectural (information and decentralization constraints). The key idea is that credit mechanisms must be simultaneously (i) statistically reliable, (ii) behaviorally robust under strategic coupling, and (iii) architecturally implementable under partial observability and decentralized execution.

*a) Counterfactual:* A counterfactual view of credit assignment estimates a component's contribution by comparing the realized system outcome to a minimally intervened alternative in which that component is removed, bypassed, or replaced, while holding the remaining pipeline fixed as much as possible [135], [136], [137], [138], [139]. In LLM-based multi-agent systems, such interventions are often implementable as trace-level edits (e.g., disabling an agent, skipping a verification stage, or masking a tool call), making counterfactual credit a practical tool for both debugging and system design. For example, SheetMind [140] reports that ablating role agents (Manager/Reflection) substantially degrades spreadsheet task success, attributing gains to decomposition and validation rather than to the action generator alone. SafeScientist [132] similarly uses module ablations over its multi-stage safety pipeline (e.g., removing prompt monitoring or discussion/tool defenses) to quantify marginal improvements in safety under adversarial prompts and malicious collaborators. MAT-Agent [141] applies the same logic by ablating adaptive agents that control augmentation/optimizer/scheduler/loss, illustrating how counterfactuals expose both marginal contributions and interaction effects among coupled modules.

*b) Coalition attribution:* Coalition attribution assigns credit by treating agents, tools, or workflow components as players in a cooperative game and measuring each player's *average marginal contribution* across all coalitions [142], [143], [144]. Concretely, given a value function $v(S)$ that evaluates the system when only a subset of components $S$ is enabled, Shapley values distribute the total outcome into per-component credits while accounting for interaction effects and redundancy:

$$\phi_{a_i} = \sum_{S \subseteq N \setminus \{a_i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \times \big[v(S \cup \{a_i\}) - v(S)\big] \tag{13}$$

This is particularly attractive in LLM agentic workflows where naive ablations can miss synergy (e.g., planning helps only when action is available) and where

multiple tools/roles may be partially substitutable. ShapleyFlow [145] applies this idea to agentic workflows by exhaustively evaluating component configurations and using Shapley values to attribute the contribution of workflow modules (e.g., planning, reasoning, action, reflection) and to discover task-specific optimal configurations. For DAG-structured multi-agent pipelines, HiveMind [146] proposes DAG-Shapley, which exploits the workflow DAG to prune non-viable coalitions and reuse intermediate outputs, enabling practical Shapley-based credit for online prompt optimization of underperforming agents. At the tool level, AgentSHAP [147] provides a model-agnostic Monte Carlo Shapley estimator that attributes an agent's final response to individual tools by replaying the agent under tool subsets and scoring outcome similarity, supporting debugging and tool pruning. Finally, Shapley-Coop [148] extends Shapley-based credit into open-ended cooperation by using marginal-contribution reasoning to guide negotiation and reward redistribution among self-interested agents, linking attribution to incentive alignment rather than only post hoc explanation.

*c) Structured evaluators / critics:* When counterfactual replays are costly or insufficiently controlled, many LLM-based multi-agent systems instead implement *structured evaluators* (critics) that score intermediate traces and gate downstream actions [149], [103], [150], [151]. These critics operationalize credit as *stage-wise, interface-aligned feedback*: they assess plan/answer quality, tool-use validity, safety or constraint compliance, and then trigger revision, re-query, or rerouting, effectively turning attribution into a monitoring-and-selection problem. In MAE [152], the Judge instantiates an explicit LLM-as-a-judge critic that scores both proposed questions and candidate answers under strict rubrics and supplies reward signals and filtering for co-evolution. MAR similarly replaces self-reflection with multiple persona critics and a judge aggregator that synthesizes critiques into a single actionable reflection, reducing confirmation bias and improving iterative correction [153]. In embodied settings, SAMALM deploys a two-tier critic architecture (global and local critics) to verify low-level robot control actions, fusing scores to decide whether to execute or re-query with critic feedback [154]. Beyond per-trajectory scoring, ARG-DESIGNER [117] uses empirical success indicators over generated collaboration graphs to supervise a topology generator and then fine-tunes for communication efficiency, effectively treating topology quality as a learned evaluation signal guiding system construction.

*d) Role-anchored:* A complementary way to make credit assignment well-defined in LLM-based multi-agent systems is to *anchor responsibility to explicit roles and protocols*. Rather than estimating marginal causal effects (counterfactual) or relying on a standalone scoring module (critic), this approach designs the system such that intermediate artifacts are produced under a fixed interaction protocol, making it clear *which role is accountable for which artifact and decision*. For example, MetaGPT [44] exemplifies this idea by encoding software-engineering SOPs into role-specialized agents (e.g., product, architecture, implementation, testing/review), so improvements and failures can be traced to role-specific deliverables under a prescribed workflow. Leveraging LLMs as Meta-Judges further structures evaluation by introducing a three-stage meta-judge pipeline (rubric design, multi-agent scoring, threshold selection), turning "judge reliability" into a protocol-governed artifact selection problem with explicit decision points [156]. At the system-construction level, MasRouter [155] defines Multi-Agent System Routing as a staged protocol (collaboration mode determination → role allocation → LLM routing), which anchors credit/cost accounting to concrete routing decisions about topology, roles, and model backbones.

### C. Empirical and theoretical insights

*a) Empirical studies:* Across recent Shapley-based credit assignment work, a consistent empirical pattern is that *faithful attribution becomes feasible only once the system is instrumented for repeatable coalition evaluation*. ShapleyFlow operationalizes this by exhaustively testing all component configurations in a controlled workflow template (e.g., Planning/Reasoning/Action/Reflection), enabling stable cross-domain comparisons and revealing that contribution is *highly task-dependent*—for instance, Action upgrades dominate in precision-critical computation tasks (Math/ATP), while Reasoning (often with Planning) dominates in interactive control tasks (OS-/RobotCoop) [142]. HiveMind reports a complementary finding for DAG-structured agent teams: naive Shapley computation is prohibitive at runtime, but exploiting structural constraints (pruning non-viable coalitions and reusing intermediate outputs) can preserve attribution exactness while reducing LLM calls by over 80%, making contribution-guided optimization practical in an online loop [146]. At a different granularity, AgentSHAP shows that Monte Carlo Shapley estimation can yield *run-to-run stable* tool-importance scores (high cosine similarity across repeats) and that removing high-attribution tools produces substantially larger quality drops than removing low-attribution tools, providing a faithfulness check for tool-level explanations and pruning decisions [147]. Finally, Shapley-Coop provides evidence that contribution estimates can support *incentive-compatible cooperation*

TABLE IV
COMPARISON OF CREDIT ASSIGNMENT STRATEGIES IN LLM-BASED MULTI-AGENT SYSTEMS. EACH STRATEGY IS ILLUSTRATED WITH TWO REPRESENTATIVE PAPERS, EVALUATED ALONG THREE KEY DIMENSIONS.

| Strategy | Core Idea | Example | Attribution Precision | Computational Overhead | Real-time Feasibility |
|---|---|---|---|---|---|
| Counterfactual | Estimate contribution via minimal interventions | SheetMind [140] | **Medium**: identifies key roles but misses synergy | **Low**: single ablation per agent | **No**: requires full task re-execution |
| | | SafeScientist [132] | **High**: identifies key modules and their synergy | **Low**: single ablation per module | **No**: offline replay only |
| Coalition Attribution | Use cooperative game theory to quantify credit assignment. | ShapleyFlow [145] | **High**: exact Shapley over all component subsets | **Extremely high**: evaluates $2^n$ coalition configurations | **No**: offline workflow analysis only |
| | | AgentSHAP [147] | **High**: captures marginal contribution and distinguishes relevant tools | **Medium**: scales linearly with number of tools | **No**: requires multiple agent replays |
| Structured Critics | Use dedicated critics to turn credit assignment into real-time monitoring. | MAE [152] | **High**: Use LLM-as-a-judge to get comprehensive scores | **Medium**: requires one additional LLM call | **Yes**: fully integrated into training loop |
| | | MAR [153] | **High**: multi-critic multi-angle analysis | **High**: 3–4 extra critic agents | **Yes**: integrated into iterative reasoning loop |
| Role-Anchored | Anchor credit to predefined roles and protocols for built-in traceability. | MetaGPT [44] | **Medium**: credit anchored to roles by design, but cannot capture dynamic contribution | **Low**: no extra evaluation or sampling | **Yes**: supports immediate traceability |
| | | MasRouter [155] | **High**: enable traceability per agent | **Low**: use lightweight controller network | **Yes**: enables integration at inference time |

in self-interested settings: Shapley-guided negotiation and redistribution improves coordination outcomes and aligns payouts more closely with measured marginal contributions than heuristic negotiation alone [148]. Taken together, these studies suggest that Shapley-style attribution is not merely post hoc explanation: when paired with appropriate instrumentation (replays, DAG constraints, or sampling), it yields actionable signals for optimizing workflows, pruning tools, and stabilizing cooperation.

*b) Decision rules:* Shapley values become operational only when coupled with *decision rules* that translate credits into interventions on prompts, topology, routing, or incentives. A first family is *bottleneck targeting*: select the lowest-contribution agent/component and update it (e.g., via reflection-based prompt optimization), while leaving high-contribution parts unchanged to avoid destabilizing the team; HiveMind instantiates this as a thresholded update trigger ($\phi_{a^*}(t) < \tau$) and iteratively refines the identified bottleneck agent's prompt [146]. A second family is *configuration selection under budget*: use Shapley-derived rankings to choose a task-specific subset of components or stronger model implementations for only the most valuable modules, as in ShapleyFlow's optimal configuration discovery over component subsets [142]. A third family is *pruning and auditing*: remove tools/agents/edges with consistently near-zero attribution (null-player behavior), and validate pruning by measuring quality drop when removing high- versus low-attribution units; AgentSHAP explicitly motivates this as tool pruning and debugging for tool-using agents [147]. Finally, in settings with strategic agents, attribution can define *redistribution rules*: allocate global surplus according to Shapley-based marginal contributions to align incentives and deter free-riding, as operationalized by Shapley-Coop's pricing and post-task reward redistribution [148]. These rules highlight a common design trade-off: attribution methods provide *estimates*, but system behavior is governed by (i) how aggressively credits are acted upon (thresholds, step sizes, pruning budgets), and (ii) what failure mode is prioritized (performance degradation, cost overruns, or incentive misalignment).

## V. RQ3.SCALABILITY: FROM SMALL GROUPS TO COMPLEX DIMENSIONS

While effective coordination and credit assignment are solvable in small, static groups, real-world applications demand systems capable of handling massive concurrency, diverse task distributions, and evolving complexity. Scaling LLM-based multi-agent systems is not merely a matter of adding more agents; it introduces non-linear dynamics where communication overhead, cognitive load, and coordination costs can rapidly outpace

TABLE V

A MATRIX VIEW OF CREDIT ASSIGNMENT IN LLM-BASED MULTI-AGENT SYSTEMS. COLUMNS DEFINE THE ATTRIBUTION UNIT (GRANULARITY). THE TOP BLOCK INSTANTIATES *credit* AS OBSERVABLE SIGNALS; THE BOTTOM BLOCK LISTS MECHANISM FAMILIES AS TYPICAL OPERATIONS. SYMBOLS INDICATE WHETHER A PAIRING IS COMMONLY USED/NATURAL (✓) OR GENERALLY ILL-SUITED (×).

| Category | Agent / Role | Turn / Message | Tool call | Artifact |
|---|---|---|---|---|
| **Credit notions (RQ2.1): observable signals** | | | | |
| **Outcome credit** | ✓ per-agent win/score; role deliverable pass | × | ✓ tool execution success; API return validity | ✓ tests pass; citation verified / patch accepted |
| **Process credit** | ✓ token/cost per agent; latency per role | ✓ rounds to converge; redundancy per turn | ✓ tool-call count; latency / retry rate | × |
| **Auditable credit** | × | ✓ claim–evidence links; critique helpfulness | × | ✓ structured logs; reproducible plans / traces |
| **Safety / compliance credit** | ✓ per-agent violation/leakage rate | ✓ unsafe content flags; policy adherence per turn | ✓ unsafe tool-use attempts; permission checks | ✓ compliant reports; redaction quality |
| **Assignment mechanisms (RQ2.2): typical operations** | | | | |
| **Counterfactual replay** | ✓ disable/swap agent; rerun pipeline | ✓ delete/replace a turn; replay suffix | ✓ block a tool; mask tool result | ✓ remove/replace artifact; rerun downstream |
| **Structured evaluators / critics** | ✓ rank agents/roles; select best output | ✓ score message; gate revise / reroute | ✓ validate schema; block unsafe calls | ✓ grade plan/code/citations; accept/reject |
| **Role-/protocol-anchored** | ✓ SOP responsibilities; role accountability | ✓ fixed debate/jury protocol; structured rounds | × | ✓ required deliverables; stage checklists |
| **Coalition attribution** | ✓ agent/role Shapley credit for $O_{\text{sys}}$ | × | ✓ tool importance via subset replay | ✓ component/artifact Shapley over configurations |

performance gains. As systems expand from simple dyads to complex organizations, the challenge shifts from enabling interaction to optimizing the "economy of scale" in intelligence. To understand how to navigate this complexity without diminishing returns, we structure our analysis around three critical inquiries:

- **RQ3.1: What are the objectives of scalability in collaboration?**
  We investigate the regularities and anti-regularities that define the boundaries of scalable collaboration. Specifically, we analyze what systematic patterns emerge in performance-vs-scale curves, communication load, and division of labor as the number of agents and task density increase [157].
- **RQ3.2: What challenges limit collaboration efficiency as scale increases?**
  We examine when and where saturation and negative collaboration emerge. This involves identifying conditions under which communication bottlenecks, interference, congestion, or over-coordination cause collaboration to saturate or become harmful to system performance [158].
- **RQ3.3: How to maximize the net utility of collaboration scaling?**
  We explore which structural priors and abstractions support migration across scales and distributions. The focus is on understanding how role abstractions, modular architectures, and specific communication protocols allow collaboration patterns learned at small scales to transfer effectively when the system is scaled up.

### A. *What are the objectives of scalability in collaboration?*

*a) Capacity:* One axis of scaling is the *capacity* of each agent: how large and capable individual policies must be for collaboration to remain reliable as the number of agents, tasks, and environments grows. In modern settings, capacity often tracks model size, context length, and compute budget for neural or LLM-based agents; in classical MAS it reflects sensing, memory, and onboard computation limits. Higher-capacity agents can, in principle, represent richer coordination strategies, better model teammates and opponents, and reason over longer horizons, thereby reducing approximation error in joint policies and enabling more flexible communication, role assignment, and reciprocity [159], [160], [161]. The same scaling, however, increases training cost, inference latency, and risk of overfitting to specific team compositions or scenarios, so that collaboration may

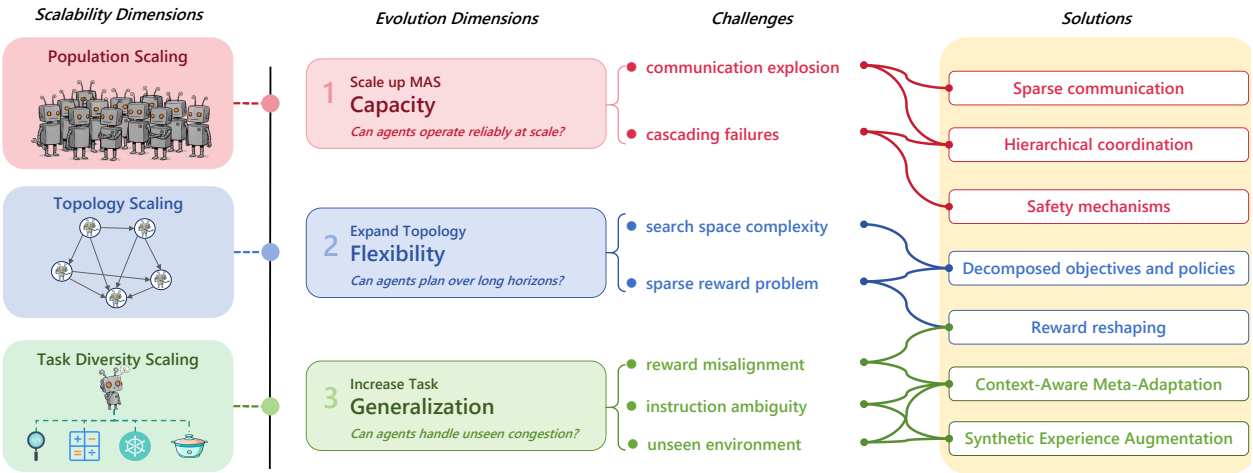


Fig. 5. A taxonomy of scalability challenges, strategies, and objectives in multi-agent systems

become more brittle when agents or tasks change [162], [163].

Representative domains highlight these trade-offs. In *multi-robot teams*, more compute per robot enables stronger perception and local planning, which supports decentralized formation control, coverage, and task allocation with less central coordination [164], [165], [166]. In *networked control and cyber-physical systems*, agents with sufficient compute can run distributed MPC or learning-based controllers that account for neighbors' dynamics, improving stability and performance in large formations, grids, or traffic networks [167], [168], [169], [170].

For *LLM-based multi-agent systems*, capacity scaling is especially visible: replacing smaller models with larger foundation models as individual agents often yields more reliable reasoning, tool use, and negotiation, improving coordination in open-ended tasks [171], [123], [172], [173]. But frontier-scale models are too slow and costly to assign to every agent in large populations. Practical designs therefore use heterogeneous architectures (e.g., a few high-capacity planner/coordinator agents plus many lighter workers) or dynamic escalation to larger models only when simpler policies fail [155], [174], [85], [175]. Overall, capacity acts as a bottleneck and an enabler: sufficient per-agent capability is necessary for robust large-scale collaboration, but scaling it up directly competes with the computational, energetic, and economic constraints of operating many agents simultaneously.

*b) Complexity:* A second axis of scaling is the *complexity* of the system: as we add agents, tasks, and communication links, the interaction topology becomes richer and the relevant planning horizon grows. Even if individual agents are unchanged, larger and more connected teams create longer chains of influence and many more joint configurations, so that local, short-sighted policies are less likely to align with system-level goals. The key question is whether agents can still form useful plans when consequences of actions unfold over many hops and time steps instead of within a small, static neighborhood.

On the structural side, expanding topology means moving from small, fixed graphs to large, possibly time-varying networks. Convergence rates and stability of coordination (e.g., consensus, synchronization, distributed optimization) then depend sensitively on graph size, diameter, and connectivity. Long paths and loops imply that perturbations can propagate widely and only manifest much later, making naive myopic controllers prone to oscillations or cascading failures. Addressing this often requires hierarchical organization or graph-aware planning that explicitly approximates these extended spatial-temporal dependencies [176], [177].

Examples illustrate this scaling of complexity. In *traffic networks*, extending from a few intersections to city-scale control introduces long congestion waves and route interdependencies; controllers must consider multi-intersection, multi-step effects, not just local queues [178]. In *power grids and infrastructure*, growing network size and tighter coupling make local setpoints capable of triggering distant overloads after several time steps, so operators and agents must reason over network-wide contingencies rather than one-step local responses [179], [180]. For *LLM-based multi-agent systems*, scaling from a small group chat to many agents interacting over long sessions creates deep dependency chains in the dialogue graph: early misunderstandings or misaligned plans can quietly propagate and compound. Without

explicit scaffolding such as hierarchical roles, graph-structured communication, or shared external memory, agents tend to default to short-horizon, locally coherent behavior that fails to yield globally coherent outcomes in these complex settings [26], [181], [27], [98].

*c) Generalization:* A third axis of scaling concerns *generalization*: as we increase task density, diversity, and load, can agents still collaborate effectively under patterns of congestion and interaction they have never seen before? In small, carefully curated settings, agents may overfit to a narrow set of traffic patterns, team compositions, or workload regimes; collaboration appears robust, but fails once the system is exposed to new arrival processes, failure modes, or strategic behaviors. At larger scales, such novelty is unavoidable: higher task density creates emergent bottlenecks, resource conflicts, and queuing effects that were rare or absent during training. The core question is whether agents learn *structures* and *principles* of coordination that transfer across loads and task mixes, rather than memorizing specific scenarios [182], [183], [184].

This issue is pronounced in domains where congestion patterns change with scale. In *traffic, mobility, and logistics*, policies tuned for moderate demand may break down under rush-hour or disruption-induced surges, producing gridlock, instability, or unfair service, because agents have never experienced these extreme regimes during training [185] . In *networks and cloud systems*, scheduling and routing strategies that work at low utilization can trigger congestion collapse, head-of-line blocking, or starvation when job sizes, tenant mixes, or attack patterns shift [186], [187], [188].

LLM-based multi-agent systems face similar generalization challenges as we densify tasks and interactions. When many agents share tools, APIs, or limited budgets, previously rare conflicts,including rate limits, queueing delays, or subtle resource contention, become common. Agents trained or tuned on isolated tasks may not know how to back off, renegotiate, or re-plan when tools are overloaded or when other agents behave in unanticipated ways [189], [190], [191], [192]. Robust collaboration at scale therefore requires training and evaluation regimes that expose agents to diverse, stressed conditions, as well as architectures (e.g., hierarchical controllers, meta-learners, or safety layers) that can adapt gracefully when congestion and interaction patterns move far beyond the training distribution [193], [194], [195], [196].

### *B. What challenges limit collaboration efficiency as scale increases?*

*a) Capacity-related challenges:* Even when we successfully scale up individual agents, their increased capability interacts with system size in problematic ways. Higher-capacity agents tend to communicate more, take more impactful actions, and require more computation per decision, so naive scaling can quickly erode any efficiency gains.

- **Communication explosion.** As we increase the number or capacity of agents, the potential volume of messages and interactions grows super-linearly: more capable agents tend to exchange richer state, beliefs, and plans, and dense topologies allow many more pairwise or group communications per time step. Without strict protocol design, this leads to a "communication explosion" that saturates bandwidth, overwhelms receivers, and makes it harder for agents to discern which signals are actually relevant for decision-making. In practice, systems must introduce compression, sparsity, or role-based communication to keep per-agent load manageable and avoid drowning high-capacity policies in low-value messages [197], [116], [117].
- **Cascading failures.** Higher-capacity agents can take more impactful actions, but this also means local errors or mis-coordination can propagate more widely and quickly. When many strong agents share resources (e.g., links, servers, markets), aggressive behaviors or mis-specified objectives by a few can trigger overloads and cascading failures that degrade collaboration for all. At scale, these cascades become harder to predict and contain, because the combined effect of many high-capacity agents interacting through shared infrastructure can exhibit sharp phase transitions between smooth operation and congestion or collapse [198], [199], [195]. Designing collaboration protocols thus requires not only enhancing individual capability, but also adding damping, throttling, or safety layers that limit the damage any single agent can cause.

*b) Complexity-related challenges:* As interaction topologies and horizons expand, agents face qualitatively harder decision problems. Long chains of influence, large graphs, and delayed effects make it difficult to assign credit, maintain long-term objectives, and search over joint strategies, even if individual agent capacity is high.

- **Sparse rewards.** In large, complex systems, the consequences of local actions often materialize only after many steps and through indirect paths. This makes feedback signals sparse and delayed: most individual actions seem to have little immediate effect, and useful gradients for coordination become extremely noisy. Learning effective collaborative policies under such conditions is slow and brittle,

and agents are prone to converging on short-sighted heuristics that happen to work on the training scenarios [200], [201].

- **Search space complexity.** Expanding topology and longer horizons also blow up the underlying search space: there are many more possible joint configurations, communication patterns, and contingency plans. Standard learning or planning methods must navigate enormous combinatorial spaces, where small policy changes can have highly non-local effects. This leads to high variance learning dynamics, brittle convergence, and strong dependence on initialization and exploration strategy [161], [202], [203]. To remain tractable, large-scale collaborative systems typically need structural assumptions—such as locality, hierarchy, or decomposability—that effectively reduce the search space while preserving enough flexibility for coordination.

*c) Generalization-related challenges:* As task diversity increases, multi-agent systems face systematic distributional shift: new task structures, new difficulty regimes (longer horizons, tighter constraints), new modalities/observation noise, and new partner behaviors appear that were absent or rare during training. The key question is not merely whether the system can solve a fixed set of benchmarks, but whether it can *transfer* its coordination machinery and accumulated experience to novel tasks without destabilizing the multi-step collaboration workflow.

- **Instruction Ambiguity.** Scaling task diversity introduces inherent vagueness in how goals are communicated. As instructions become more complex or abstract, the mapping from natural language to precise multi-agent coordination becomes underspecified. This ambiguity often leads to "semantic drift", where different agents interpret the same prompt through divergent internal biases, resulting in misaligned decomposition and conflicting sub-task execution. Without robust grounding, the system fails to resolve linguistic nuances, leading to coordination breakdowns even when individual agents are capable [35].
- **Reward Mismatch.** Across heterogeneous tasks, defining a universal "success" metric is challenging as optimal behavior varies by domain: prioritizing safety in one context versus efficiency in another. When a unified reward model or proxy verifier is applied across diverse tasks, rewards often become non-comparable or poorly calibrated. This mismatch facilitates "reward hacking", where agents over-optimize for superficial format or easy-to-attain metrics at the expense of true task completion. Such inconsistencies complicate credit assignment in multi-step pipelines and hinder stable cross-task scaling [182], [183].
- **Unseen Environments.** Generalization is further tested by structural shifts in the environment, such as novel observation noise, shifting transition dynamics, or unfamiliar partner policies. In these unseen settings, pre-trained collaboration patterns often prove brittle; an agent's inability to adapt to out-of-distribution (OOD) environmental feedback can trigger cascading failures across the entire pipeline. The challenge lies in maintaining "workflow resilience"—ensuring that the system-level interaction remains stable and adaptive when the underlying environment no longer matches the training distribution [204], [182], [183].

### C. How to adapt to complex collaboration paradigm?

The previous subsection outlined major challenges that arise when scaling collaboration along capacity, complexity, and generalization. Scaling up agents and environments can unlock new capabilities, but it also introduces communication bottlenecks, instability, and brittleness under distribution shift. We ask how to design systems so that the benefits of scaling can be maximized.

*a) Sparse communication:* One way to control both communication explosion and search space complexity is to explicitly constrain who talks to whom, and how often. Instead of fully connected or dense communication graphs, large-scale systems can use sparse topologies with bounded degree, locality constraints, or dynamically pruned neighborhoods, possibly implemented via selective communication [88], [73] or learnable routing over a graph structure [205], [115], [116], [117]. Notably, recent approaches have diverged in their optimization of these structures: methods such as AgentDropout [115] and AgentPrune [116] employ pruning techniques to dynamically eliminate redundant connections, whereas frameworks like Assemble [117] focus on training directly upon sparse configurations to generate intrinsically low-overhead topologies. This keeps per-agent bandwidth and cognitive load roughly constant as the system grows, and reduces the combinatorial number of possible joint communication patterns that learning algorithms must explore. In practice, such pruning can be combined with lossy compression, message summarization, or periodic rather than continuous exchange to further curb communication volume.

*b) Hierarchical coordination structures:* Beyond sparsifying links, many systems benefit from adding explicit organizational structure: roles, teams, and hierarchies. Instead of all agents coordinating symmetrically at

the same level, some agents can aggregate information and make higher-level decisions, while others focus on local control. This reduces the need for global synchronization, limits cascade lengths (by containing failures within subgroups), and allows fast local reactions without always invoking slow, global reasoning [206], [47]. Hierarchical controllers, manager–worker patterns, or regional coordinators can also shrink the effective search space by constraining which joint behaviors are even considered. For instance, the HALO framework [207] implements such stratification through a Hierarchical Reasoning Stack that decouples high-level planning from low-level execution. Specifically, it employs a top-tier agent for stepwise task decomposition and mid-level agents for dynamic role instantiation, while utilizing a Monte Carlo Tree Search (MCTS) engine to optimize execution workflows at the inference layer. This design not only mitigates the cognitive overload associated with monolithic agents but also enhances adaptability by treating multi-agent collaboration as a structured state-space search problem.

*c) Resource guardrails:* To prevent cascading failures and myopic overuse of shared resources, large-scale collaborative systems often need explicit constraints on the impact of any single agent or small group. Mechanisms such as quotas [73], [208], rate limiting [93], or safety envelopes [209], [210] can keep utilization within safe bounds even when some agents behave aggressively or are mis-specified. Inspired by frameworks such as GuardAgent [211], modern guardrails are evolving from static counting mechanisms into intelligent, reasoning-based monitors. Rather than merely tracking request frequencies, these agents dynamically analyze the context of proposed action plans and evaluate them against the global system state to verify safety. Additionally, carefully designed fail-safe defaults—e.g., conservative fallback policies when uncertainty or load is high—can dampen feedback loops that would otherwise amplify small errors into system-wide crises. These mechanisms do not improve individual planning per se, but they shape the feasible action space so that local, possibly myopic optimization is less likely to cause global damage.

*d) Reward reshaping:* Addressing sparse and delayed rewards, as well as myopic optimization, often requires improving the learning signal itself. Reward shaping introduces intermediate, more informative feedback that correlates with long-term goals, while still preserving the optimal policies under the original objective when done carefully [201]. In multi-agent settings, counterfactual credit assignment methods [138], value decomposition [212], and influence-based rewards [200], [213] can help agents understand how their local actions contribute to global outcomes, even when the final metric is only observed infrequently. These tools make it more feasible for agents to optimize genuinely long-horizon objectives instead of falling back to short-sighted proxies simply because those are the only signals they can reliably learn from.

*e) Decomposed objectives:* Another way to tame search space complexity and improve credit assignment is to decompose the global problem into smaller, more structured subproblems. This can mean factorizing the joint value function or policy over subsets of agents, exploiting task locality, or using explicit task/goal decomposition so that different agents or modules handle different sub-tasks [214], [215]. Such factorization reduces the combinatorial explosion of joint action spaces and can enable more stable learning dynamics, while still allowing coordination through carefully designed coupling terms or shared constraints. Proper decomposition also helps align local objectives with global ones, mitigating myopic behavior by making long-term structure visible at the subproblem level.

*f) Context-aware meta-adaptation:* To navigate complex and non-stationary environments, LLM agents must transcend static execution by employing context-aware meta-adaptation. This mechanism enables agents to perceive task-specific nuances and internal cognitive cues—such as confidence or uncertainty—to dynamically recalibrate their reasoning trajectories and system configurations. For example, MPDF [216] introduces a meta-cognitive layer where agents evaluate their internal states to strategically choose between Persisting, Refining, or Conceding. In a similar vein, ReMA [217] utilizes MARL to elicit high-level strategic oversight, enabling a dedicated agent to monitor and rectify the reasoning process of its subordinates. Beyond algorithmic flexibility, this adaptation facilitates rapid domain transfer; AdaptAgent [218] leverages few-shot human demonstrations as contextual anchors to adapt multimodal agents to unseen web interfaces. Furthermore, MemEvolve [219] pushes the boundary of self-evolution by meta-adapting the memory architecture itself, tailoring storage and retrieval schemas to the specific affordances of diverse domains like software engineering or scientific discovery. By integrating these context-aware meta-regulatory processes, agentic systems can mitigate "meta-cognitive blindspots", fostering unprecedented robustness and continuous self-improvement in open-ended task solving.

*g) Synthetic experience augmentation:* To overcome the limited generalization of LLM agents in unseen or open-ended environments, synthetic experience augmentation serves as a critical strategy to expand the training distribution. By generating high-fidelity, sim-

ulated trajectories, this approach exposes agents to a vast array of edge cases and task variations that are often absent in human-labeled datasets. For instance, AgentGen [220] enhances out-of-distribution robustness by automatically synthesizing diverse environments and tasks with a curriculum-based difficulty scale. Similarly, AgentTuning [221] leverages the synthetic "AgentInstruct" dataset to bridge the performance gap between open-source models and proprietary systems across various agentic benchmarks. To ensure social and behavioral generalization, MATRIX [222] utilizes multi-agent simulations to generate realistic, human-centric interaction scenarios. Furthermore, AgentSGEN [223] automates the creation of semantically consistent 3D scenes for spatial reasoning, while MAS-GPT [224] treats the configuration of multi-agent architectures as a generative task to foster structural adaptability. By populating the learning space with these diverse, model-generated experiences, agentic systems can achieve superior zero-shot transfer and maintain robust performance in complex, real-world applications.

### D. Empirical and theoretical insights

*a) Empirical studies:* Across cooperative control, networked systems, and LLM-based multi-agent pipelines, empirical studies report several recurring scaling patterns. First, *saturation* is common: performance improves rapidly when moving from a non-collaborative baseline to limited collaboration, then exhibits diminishing marginal returns as population size, message rate, or connectivity increases; beyond a regime-dependent threshold, additional agents or communication can yield negligible gains or even *negative collaboration*. Second, multiple bottlenecks emerge with scale. Communication frequently grows super-linearly (dense graphs, repeated deliberation rounds), leading to bandwidth/latency blow-ups and a rising fraction of low-value messages; in LLM-based systems this appears as token and round explosion with compounding inference latency. Third, scale increases the risk of *cascading failures*: local errors, mis-specified subgoals, or resource contention can propagate through long dependency chains, producing sharp phase transitions from stable operation to congestion or collapse. Fourth, generalization degrades as task density and heterogeneity increase: policies overfit to particular team sizes, partner distributions, or load regimes and fail under partner swap, agent churn, or domain shift; this is amplified by curriculum gaps where training occurs in simplified settings and deployment operates in high-contention regimes. Finally, empirical evidence suggests that *structure helps*: sparse connectivity, hierarchical coordination, modular roles, and explicit throttling/rate-limits often preserve or improve net utility at scale by bounding interaction complexity and containing failure propagation.

*b) Insights and decision rules:* These findings support several decision-oriented rules for scalable collaboration.

- *Optimize for frontiers, not single points*: at scale, report and select designs by performance–cost frontiers (return/variance vs. bits/latency/energy/compute) and identify saturation/negative-collaboration thresholds; favor architectures that maintain a positive marginal value of added agents/links.
- *Bound interaction complexity*: when communication or coordination overhead grows faster than linearly, introduce sparsity (bounded-degree graphs, selective routing), temporal gating (communicate/debate only when necessary), and summarization/compression; prefer environment-mediated coordination when explicit messaging is costly.
- *Contain cascades with hierarchy and throttling*: for tightly coupled systems with shared resources, adopt hierarchical control (manager–worker, regional coordinators), quotas/rate limits, and conservative fallback policies to dampen feedback loops; evaluate with stress tests that measure degradation curves and recovery latency under overload, dropouts, and tool/API volatility.
- *Plan for migration*: if agent churn, partner change, or domain shift is expected, prioritize transferable abstractions (role/skill modules, modular policies, graph-invariant representations) and training/evaluation across varying team sizes and partner distributions; report cross-scale transfer curves and robustness to composition shift.
- *Choose the scaling lever explicitly*: when performance collapses with larger $n$, diagnose whether the binding constraint is *capacity* (insufficient per-agent competence), *complexity* (long-horizon/topological coupling), or *generalization* (shift in load/task mix), and select interventions accordingly (heterogeneous capacity allocation, hierarchical planning, scaling-aware robust training).

## VI. RQ4. Evaluation Protocols Aligned with Decisions

As multi-agent systems evolve from simple conversational partners to complex problem-solving units, traditional evaluation paradigms centered on single-agent capabilities or static text generation quality become insufficient. Evaluating collaboration requires looking

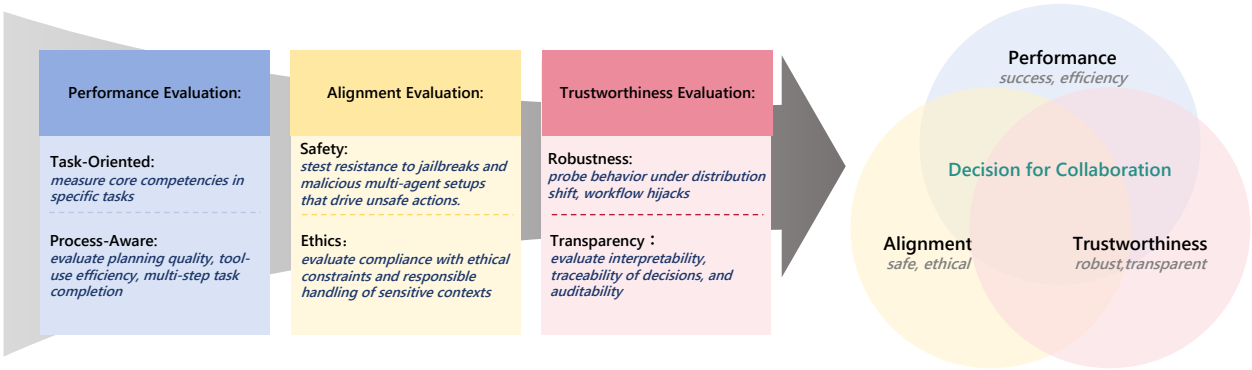


Fig. 6. Dimensions of existing evaluation tasks versus the objectives of decision-oriented evaluation.

beyond *what* an agent says to *how* the group functions as a cohesive entity. It necessitates disentangling individual contributions from collective outcomes and verifying whether the collaborative process genuinely aligns with the intended decision-making goals. To address the lack of standardized assessment in this emerging field, we critically review existing methodologies and propose directions for more robust frameworks through two key questions:

- **RQ4.1: What are the metrics and benchmarks currently employed to evaluate collaboration?**
  We provide a comprehensive survey of the current landscape of assessment tools. This includes categorizing existing benchmarks based on task types (e.g., social simulation, game theory, software engineering) and analyzing the metrics used to quantify success—ranging from outcome-oriented metrics (success rate, efficiency) to process-oriented metrics (communication cost, consensus capability, and social behaviors).
- **RQ4.2: How can we establish a decision-aligned evaluation framework?**
  We identify the gaps between current metrics and actual collaborative utility. We discuss the necessity of moving towards "decision-aligned" evaluation, where metrics prioritize the quality, safety, and robustness of the final decision rather than merely the fluency of interaction. This involves exploring dynamic evaluation environments, human-in-the-loop assessments, and the alignment of agent incentives with system-level objectives.

### *A. What are the metrics and benchmarks currently employed to evaluate collaboration?*

In collaborative systems, evaluation is not reducible to a single task score but rather a collection of measurements aligned with distinct deployment decisions. Concretely, we need to assess (i) whether the system can achieve its objectives (*capability and helpfulness*), (ii) whether its decisions and actions comply with normative constraints (*safety and ethics*), and (iii) whether it remains operationally dependable and governable in the face of uncertainty and change (*trustworthiness*). Accordingly, existing metrics and benchmarks for collaboration can be grouped into three categories: *Performance Evaluation* (capability & helpfulness), *Alignment Evaluation* (safety & ethics), and *Trustworthiness Evaluation* (robustness, reliability & transparency). These categories address complementary questions—"does it work?", "does it behave appropriately?", and "can we depend on and govern it in the wild?"—yet they also exhibit inherent tensions: optimizing task performance can incentivize shortcutting and metric gaming, strict constraints can reduce efficiency, and trustworthiness-focused stress tests may miss the normative benefit–risk trade-offs that ultimately determine deployability. In RQ4.1, we systematize the metrics and benchmarks currently used to evaluate collaboration and highlight their coverage and blind spots.

*a) Performance evaluation: capability & helpfulness:* In collaborative agentic systems, *performance evaluation* assesses whether a team of agents can accomplish intended objectives under nominal operating assumptions. Beyond end outcomes (e.g., success rate, reward, completion time, and resource/cost consumption), performance metrics increasingly target the *competencies that produce* those outcomes in agentic workflows, including planning quality, tool-use efficiency, coordination effectiveness, and multi-step task completion. In practice, this layer is anchored by domain benchmarks: for collaborative software engineering and code generation, common task-oriented suites include HumanEval and MBPP [225], [226] as well as more end-to-end issue-resolution settings such as SWE-Bench [11]; for embodied collaboration, benchmarks

| Evaluation Types | Focus on | Benchmarks/Datasets | Used by |
|---|---|---|---|
| **Performance** *Assess whether agents accomplish objectives (capability) and provide utility (helpfulness) under nominal operating assumptions.* | Task-oriented | HumanEval [225] | [44], [235], [236] |
| | | MBPP [226] | [155], [237], [238] |
| | | SWE-Bench [11] | [10], [239], [240] |
| | | RoCoBench [227] | [227], [129], [241] |
| | | C-WAH [49] | [49], [242], [47] |
| | | TDW-MAT [228] | [242], [243], [244] |
| | | Minecraft [229] | [245], [246], [247] |
| | | MMLU [230] | [205], [115], [116] |
| | | MedQA [231] | [248], [249], [250] |
| | | PubMedQA [232] | [248], [251], [252] |
| | | GSM8K [233] | [115], [116], [117] |
| | | StrategyQA [234] | [253], [254], [255] |
| | Process-aware | AgentBench [256] | [261], [262] |
| | | AgentBoard [257] | [263], [264] |
| | | GAIA [12] | [265], [266], [205] |
| | | MINT [258] | [267], [268], [264] |
| | | VisualWebArena [259] | [269] |
| | | T-Eval [260] | [270] |
| **Alignment** *Assess normative acceptability: adherence to safety policies, ethical principles, and refusal boundaries.* | Safety | Do-not-answer [271] | [277], [278] |
| | | DoomArena [272] | [279] |
| | | Agent-SafetyBench [273] | [273] |
| | | S-Eval [274] | - |
| | | BAD [275] | [280], [281], [282] |
| | | BeaverTails [276] | [283] |
| | Ethics | TrustfulQA [284] | [287], [288] |
| | | Ethical Agent [285] | [289] |
| | | MIST [286] | [290] |
| **Trustworthiness** *Assess operational dependability and governability under uncertainty (robustness, reliability & transparency).* | Robustness | PARTNR [291] | - |
| | | MARLSafe [292] | - |
| | | RiskAwareBench [293] | - |
| | | HAICosystem [294] | [47] |
| | | ToolEmu [295] | [297] |
| | | PromptBench [296] | [298], [299] |
| | Transparency | TRAIL [300] | - |
| | | Nemo guardrails [301] | [302] |

TABLE VI
CATEGORIZATION OF EXISTING MAS EVALUATION WORKS. WE CLASSIFY EVALUATIONS INTO THREE DIMENSIONS: **PERFORMANCE** (CAPABILITY UNDER NOMINAL CONDITIONS), **ALIGNMENT** (ADHERENCE TO NORMS/SAFETY), AND **TRUSTWORTHINESS** (DEPENDABILITY UNDER UNCERTAINTY).

such as RoCoBench, C-WAH, and TDW-MAT [227], [49], [228] are widely used; and for knowledge/reasoning evaluation, suites like MMLU, GSM8K, and StrategyQA remain standard anchors [230], [233], [234]. For tool-using agents, task suites such as AgentBench, AgentBoard, GAIA, MINT, and VisualWebArena have become common measurement substrates [256], [257], [12], [258], [259]. Mirroring the shift from passive QA to goal-driven orchestration, recent practice therefore spans (i) *task-oriented* suites that measure core competencies in well-specified collaboration tasks; (ii) *process-aware* evaluations that quantify planning/execution quality in long-horizon workflows (e.g., progress toward subgoals, tool-use accuracy, and completion of intermediate milestones), without treating auditability as the primary objective; and (iii) *user-perceived helpfulness* measures that capture real utility in interactive settings (e.g., effort reduction, usability, and satisfaction).

*b) Alignment evaluation: safety & ethics: Alignment evaluation* assesses whether collaborative agents make decisions that are *normatively acceptable*: they pursue user goals while adhering to safety policies, ethical principles, and domain constraints as autonomy and delegation increase. To avoid overlap with robustness, this category centers on *norm compliance* rather than general performance degradation under perturbations. In practice, alignment evaluation typically combines: (i) *safety compliance benchmarks* that test policy adherence

and refusal boundaries (e.g., Do-not-answer, AgentSafetyBench, S-Eval) [271], [273], [274]; (ii) *adversarial safety (security) testing* that evaluates resistance to *intentional* norm-violation attempts (e.g., jailbreaks and (in)direct prompt injection), often operationalized via red-teaming arenas and misuse datasets such as DoomArena and BeaverTails [272], [276]; and (iii) *ethics & rights compliance benchmarks* that assess fairness/non-discrimination, privacy and consent, and responsible behavior in sensitive contexts, using resources such as TruthfulQA, Ethical Agent, and misinformation-focused benchmarks [284], [285], [286]. Under this boundary, *truthfulness* belongs to alignment when framed as a normative constraint (e.g., non-deceptive communication and avoidance of fabricated justifications intended to mislead) [284], while *factual accuracy and confidence calibration* are treated as trustworthiness properties.

*c) Trustworthiness evaluation: robustness, reliability & transparency: Trustworthiness evaluation* complements alignment by assessing whether collaborative agent systems are *operationally dependable and governable* in open, uncertain, and evolving environments. This dimension foregrounds three tightly related aspects. First, *robustness*: stability of task performance and recovery under distribution shift, partial observability, tool/API volatility, long-horizon error accumulation, and cascading failures. This is increasingly probed by robustness-oriented suites for agents and interaction, such as PARTNR, MARLSafe, RiskAwareBench, ToolEmu, and PromptBench [291], [292], [293], [295], [296], including cases where injection primarily causes *workflow disruption* (task derailment) rather than norm violation. Second, *reliability*: whether the system appropriately represents and acts on uncertainty, captured by calibration, confidence signaling, abstention/ask-for-help behavior, and safe fallback strategies. Third, *transparency*: whether decisions and outcomes are traceable and auditable through structured provenance (prompts, tool calls, intermediate state, and inter-agent communication), supported by both benchmark-style instrumentation (e.g., TRAIL) and production observability/audit stacks (e.g., OpenTelemetry, LangSmith, and guardrail frameworks) [300], [272], [273], [301]. Robustness is commonly measured via stress tests that report degradation curves, worst-case performance, and recovery latency; reliability is quantified via calibration error, selective prediction/abstention quality, and fallback effectiveness; and transparency is operationalized through logging and audit tooling that enables post-hoc debugging and responsibility attribution, while introducing trade-offs with scalability, cost, and privacy.

### *B. How can we establish a decision-aligned evaluation framework?*

A decision-aligned evaluation framework should make benchmark evidence directly actionable for deployment choices. Our starting point is the net value of collaboration: a collaboration design is desirable only if it improves the system-level objective *after* accounting for collaboration costs and constraints. Using the formulation in Section 2, we evaluate designs by their expected net gain

$$\Delta \;=\; \mathbb{E}_{\tau\sim M_{\text{collab}}}[O_{\text{sys}}(\tau) - C_{\text{collab}}(\tau)] - \mathbb{E}_{\tau\sim M_{\text{base}}}[O_{\text{sys}}(\tau)]\,, \tag{14}$$

subject to deployment constraints and normative requirements. This perspective implies that reporting a single task score is insufficient: what matters is how performance changes as we vary budgets, scale, and stressors, and whether the resulting operating points satisfy alignment and trustworthiness gates.

*a) From scores to frontiers: efficiency as the primary reporting object:* For RQ1-style decisions (whether communication is worth it under budgets), we recommend reporting *efficiency frontiers*: curves (or Pareto sets) of task performance versus collaboration cost, where $C_{\text{collab}}$ is expanded into measurable budgets such as bit-rate/tokens, end-to-end latency, energy, and GPU-hours. Concretely, each method should be evaluated across a grid of budget settings (message rate/length, number of interaction rounds, tool-call limits), producing marginal-value curves (gain per additional bit/token/second) and task-dependent saturation points. This turns "algorithm A beats B" into a decision-relevant statement: "A dominates B in the feasible budget region".

*b) Gated deployability: separating objectives from constraints:* To keep the three evaluation dimensions orthogonal, we treat *performance* as the primary objective, while *alignment* and *trustworthiness* act as deployability gates. Specifically, for each operating point on the efficiency frontier, we also report: (i) alignment violation rates under policy compliance and adversarial safety tests (e.g., unsafe-action or policy-violation rate); and (ii) trustworthiness metrics capturing robustness under distribution shift and operational perturbations (degradation and recovery) together with reliability (calibration/abstention/fallback) and transparency (provenance and audit readiness). An operating point is considered deployable only if it lies within a feasible region defined by these gates (e.g., violation rates below thresholds, bounded worst-case degradation, and auditable traces available), making the benefit–risk trade-off explicit.

*c) Scale and migration curves: evidence that survives regime change:* For RQ3-style decisions, point

estimates at a fixed team size are not decision-aligned. We therefore require *scale/migration curves*: performance–cost frontiers and deployability gates evaluated as functions of (i) number of agents, (ii) task density/load, and (iii) topology/connectivity, as well as under partner churn and domain shift. These curves reveal saturation regimes, negative-collaboration thresholds, and failure boundaries, and enable "feasibility maps" that summarize where a collaboration design produces positive net value.

*d) Minimum reporting standard:* A minimal decision-aligned report should include: (1) the efficiency frontier with confidence intervals across random seeds; (2) the alignment and trustworthiness gates for the same operating points (violation rates, worst-case degradation and recovery, calibration/abstention quality, and audit/provenance availability); (3) scale/migration curves (varying $n$, load, and shift); and (4) ablations that isolate which collaboration components (communication, credit assignment, hierarchy) move the frontier and which primarily affect gates. This standard makes evidence comparable across papers and directly supports design selection.

## VII. Future Trends

*a) Adaptive workflow orchestration:* The transition from static, hand-crafted pipelines to adaptive orchestration is essential for deploying agents in variable real-world environments where "one-size-fits-all" is inefficient. The core challenge lies in the rigidity of current systems, which struggle to balance the trade-off between performance and cost; they often over-spend tokens on simple queries or fail to escalate sufficiently for complex reasoning, lacking the ability to dynamically gauge instance difficulty. To address this, future orchestrators must function as decision-theoretic optimizers that dynamically route subtasks and determine *when* to invoke collaboration based on risk and budget constraints. This involves developing policies for early stopping and adaptive resource allocation that explicitly optimize the performance–cost frontier, ensuring that additional coordination is only triggered when it yields a statistically significant improvement in outcome quality.

*b) Adaptive and dynamic evaluation paradigms:* As agents become more specialized, evaluation is shifting from a static benchmarking process to a dynamic, bidirectional relationship where models and datasets increasingly select each other. A major difficulty with current paradigms is that static benchmarks often fail to capture the specific interaction styles or inductive biases of new agents, leading to scores that do not generalize to deployment; conversely, models may overfit to fixed evaluation sets without learning robust behaviors. Future research should formalize this co-selection as an optimization problem, where practitioners curate datasets that match a target model's profile while agents actively select training data to maximize generalization per unit cost. Ultimately, the field must move toward metrics of *predictive validity*, assessing whether a chosen evaluation protocol accurately forecasts safety violations and robustness under distribution shifts for the specific agent class it serves.

*c) Acceleration and efficiency:* As collaborative systems scale, computational cost and latency are becoming binding constraints that determine feasibility rather than just secondary metrics. The fundamental hurdle is that multi-agent designs inherently amplify inference calls and sequential dependencies, creating a multiplicative overhead that can render complex reasoning pipelines impractical for real-time applications. To overcome this, research must focus on reducing the marginal cost of collaboration through techniques such as speculative execution of subtasks, structured message compression, and the caching of intermediate reasoning steps. Acceleration should be evaluated not merely as a raw speedup, but as a *frontier shift*—demonstrating the ability to achieve the same reasoning quality with significantly fewer tokens and lower latency without compromising alignment or reliability.

*d) Decentralized and heterogeneous ecosystems:* The evolution of agent systems is trending toward decentralized and heterogeneous ecosystems, reflecting real-world scenarios where agents differ in capability, ownership, and permissions. However, this shift introduces the "consensus paradox", where the system must maintain robust decision-making without the bottleneck of a central authority, all while managing the risks of cascading failures caused by weak or compromised components in a loosely coupled network. Addressing these challenges requires the development of topology-aware coordination protocols, such as sparse graphs or local neighborhoods, which bound interaction complexity. Furthermore, self-governance mechanisms must be established to resolve conflicts between agents optimizing for local efficiency versus those prioritizing global stability, ensuring the ecosystem remains resilient under churn and bounded bandwidth.

*e) Trustworthy, private, and secure interaction:* The expansion of agents into high-stakes domains such as finance and cyber-physical systems necessitates a security paradigm that goes beyond simple access control. The complexity arises from an expanded attack surface where "tool-poisoning" or a single compromised agent can trigger cascading physical failures or privacy breaches, especially when sensitive data is shared across dynamic

chains of interaction. To mitigate these risks, future systems must implement context-aware privacy designs where agents dynamically negotiate data protection levels based on task criticality and user consent. This must be paired with rigorous provenance tracking and immutable audit trails, ensuring that high-impact actions are verifiable and that liability can be accurately apportioned among developers, operators, and agents in the event of a failure.

*f) Standardization and interoperability:* For heterogeneous agents to collaborate effectively, communication protocols must evolve to support greater semantic richness and adaptability. The primary obstacle is the "versatility–efficiency–portability trilemma", where current protocols are either too rigid to handle diverse tasks or too unstructured to guarantee type safety and prevent hallucinations during complex negotiations. Future work must therefore prioritize the development of universal semantic frameworks and adaptive protocols that allow agents from different providers to align on goals and value hierarchies. By bridging the gap between natural language flexibility and structured API determinism, these standards will enable seamless interoperability, preventing the fragmentation of the agent landscape into incompatible silos.

## VIII. Conclusion

In this paper, we position the trade-off analysis of multi-agent collaboration as a critical decision-making framework for deploying efficient and scalable intelligent systems, and have presented a comprehensive survey addressing the core question: Is collaboration worth the cost? Specifically, we have structured our exploration around four fundamental research questions. First, regarding communication efficiency (RQ1), we synthesized strategies for optimizing information exchange under bandwidth constraints, highlighting the shift from continuous broadcasting to selective and compressed messaging. Second, addressing credit assignment (RQ2), we examined how to align individual incentives with global objectives, ensuring that agents learn effective policies even in sparse-reward environments. Third, we investigated the limits of scalability (RQ3), identifying the saturation points where the overhead of coordination outweighs its benefits as the number of agents grows. Finally, by reviewing evaluation methodologies (RQ4), we critiqued current benchmarks and advocated for comprehensive metrics that account for computational cost, robustness, and real-world deployability.

As multi-agent systems transition from simulation to reality, answering these questions becomes prerequisite for success. Future research must move beyond maximizing raw performance to prioritizing the efficiency-robustness frontier, ensuring that collaborative agents are not only powerful but also economically and operationally viable in complex environments.

## References


[1] Y.-C. Choi and H.-S. Ahn, "A survey on multi-agent reinforcement learning: Coordination problems," in *Proceedings of 2010 IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications*. IEEE, 2010, pp. 81–86.

[2] J. Chai, Z. Zhao, Y. Zhu, and D. Zhao, "A survey of cooperative multi-agent reinforcement learning for multi-task scenarios," *Artificial Intelligence Science and Engineering*, vol. 1, no. 2, pp. 98–121, 2025.

[3] D. Kong, S. Lin, Z. Xu, Z. Wang, M. Li, Y. Li, Y. Zhang, H. Peng, X. Chen, Z. Sha *et al.*, "A survey of llm-driven ai agent communication: Protocols, security risks, and defense countermeasures," *arXiv preprint arXiv:2506.19676*, 2025.

[4] V. Trianni, *Evolutionary swarm robotics: evolving self-organising behaviours in groups of autonomous robots*. Springer, 2008.

[5] B. Khaldi and F. Cherif, "An overview of swarm robotics: Swarm intelligence applied to multi-robotics," *International Journal of Computer Applications*, vol. 126, no. 2, pp. 31–37, 2015.

[6] K. V. R. Devi, B. Smitha, S. Lakhanpal, R. Kalra, V. A. Sethi, and S. K. Thajil, "A review: Swarm robotics: Cooperative control in multi-agent systems," in *E3S Web of Conferences*, vol. 505. EDP Sciences, 2024, p. 03013.

[7] L. Gupta, R. Jain, and G. Vaszkun, "Survey of important issues in uav communication networks," *IEEE communications surveys & tutorials*, vol. 18, no. 2, pp. 1123–1152, 2015.

[8] R. Luo, T. J. van den Boom, and B. De Schutter, "Multi-agent dynamic routing of a fleet of cybercars," *IEEE Transactions on Intelligent Transportation Systems*, vol. 19, no. 5, pp. 1340–1352, 2017.

[9] J. Wang, X. Zhang, X. He, and Y. Sun, "Bandwidth allocation and trajectory control in uav-assisted iov edge computing using multiagent reinforcement learning," *IEEE Transactions on Reliability*, vol. 72, no. 2, pp. 599–608, 2022.

[10] J. Yang, C. E. Jimenez, A. Wettig, K. Lieret, S. Yao, K. Narasimhan, and O. Press, "Swe-agent: Agent-computer interfaces enable automated software engineering," *Advances in Neural Information Processing Systems*, vol. 37, pp. 50 528–50 652, 2024.

[11] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. Narasimhan, "Swe-bench: Can language models resolve real-world github issues?" *arXiv preprint arXiv:2310.06770*, 2023.

[12] G. Mialon, C. Fourrier, T. Wolf, Y. LeCun, and T. Scialom, "Gaia: a benchmark for general ai assistants," in *The Twelfth International Conference on Learning Representations*.

[13] S. Rakshit and R. K. Guha, "Fair bandwidth sharing in distributed systems: A game-theoretic approach," *IEEE Transactions on Computers*, vol. 54, no. 11, pp. 1384–1393, 2005.

[14] W. R. Heinzelman, A. Chandrakasan, and H. Balakrishnan, "Energy-efficient communication protocol for wireless microsensor networks," in *Proceedings of the 33rd annual Hawaii international conference on system sciences*. IEEE, 2000, pp. 10–pp.

[15] N. Naderializadeh, J. J. Sydir, M. Simsek, and H. Nikopour, "Resource management in wireless networks via multi-agent deep reinforcement learning," *IEEE Transactions on Wireless Communications*, vol. 20, no. 6, pp. 3507–3523, 2021.

[16] F. Liu, T. Wei, Z. Zhang, J. Chen, and L. Lu, "Survey of opponent modeling: from game ai to combat deduction," in *2024 36th Chinese Control and Decision Conference (CCDC)*. IEEE, 2024, pp. 825–832.

[17] W. Long, T. Hou, X. Wei, S. Yan, P. Zhai, and L. Zhang, "A survey on population-based deep reinforcement learning," *Mathematics*, vol. 11, no. 10, p. 2234, 2023.

[18] Y. Yang, H. Chai, Y. Song, S. Qi, M. Wen, N. Li, J. Liao, H. Hu, J. Lin, G. Chang *et al.*, "A survey of ai agent protocols," *arXiv preprint arXiv:2504.16736*, 2025.

[19] S.-J. Chung, A. A. Paranjape, P. Dames, S. Shen, and V. Kumar, "A survey on aerial swarm robotics," *IEEE Transactions on robotics*, vol. 34, no. 4, pp. 837–855, 2018.

[20] Z. H. Ismail, N. Sariff, and E. G. Hurtado, "A survey and analysis of cooperative multi-agent robot systems: challenges and directions," *Applications of Mobile Robots*, vol. 5, pp. 8–14, 2018.

[21] I. F. Akyildiz, W. Su, Y. Sankarasubramaniam, and E. Cayirci, "A survey on sensor networks," *IEEE Communications magazine*, vol. 40, no. 8, pp. 102–114, 2002.

[22] A. Amirkhani and A. H. Barshooi, "Consensus in multi-agent systems: a review," *Artificial Intelligence Review*, vol. 55, no. 5, pp. 3897–3935, 2022.

[23] F. Jiang, C. Pan, L. Dong, K. Wang, O. A. Dobre, and M. Debbah, "From large ai models to agentic ai: A tutorial on future intelligent communications," *arXiv preprint arXiv:2505.22311*, 2025.

[24] Z. Deng, Y. Guo, C. Han, W. Ma, J. Xiong, S. Wen, and Y. Xiang, "Ai agents under threat: A survey of key security challenges and future pathways," *ACM Computing Surveys*, vol. 57, no. 7, pp. 1–36, 2025.

[25] Y. Wang, S. Guo, Y. Pan, Z. Su, F. Chen, T. H. Luan, P. Li, J. Kang, and D. Niyato, "Internet of agents: Fundamentals, applications, and challenges," *arXiv preprint arXiv:2505.07176*, 2025.

[26] T. Guo, X. Chen, Y. Wang, R. Chang, S. Pei, N. V. Chawla, O. Wiest, and X. Zhang, "Large language model based multi-agents: A survey of progress and challenges," *arXiv preprint arXiv:2402.01680*, 2024.

[27] K.-T. Tran, D. Dao, M.-D. Nguyen, Q.-V. Pham, B. O'Sullivan, and H. D. Nguyen, "Multi-agent collaboration mechanisms: A survey of llms," *arXiv preprint arXiv:2501.06322*, 2025.

[28] Y. Kim, K. Gu, C. Park, C. Park, S. Schmidgall, A. A. Heydari, Y. Yan, Z. Zhang, Y. Zhuang, M. Malhotra, P. P. Liang, H. W. Park, Y. Yang, X. Xu, Y. Du, S. Patel, T. Althoff, D. McDuff, and X. Liu, "Towards a science of scaling agent systems," 2025. [Online]. Available: https://arxiv.org/abs/2512.08296

[29] Y. Lashkari, M. Metral, and P. Maes, "Collaborative interface agents," *Readings in agents*, pp. 111–116, 1997.

[30] S. Cai, J. Zhang, K. Bao, C. Gao, Q. Wang, F. Feng, and X. He, "Agentic feedback loop modeling improves recommendation and user simulation," in *Proceedings of the 48th International ACM SIGIR conference on Research and Development in Information Retrieval*, 2025, pp. 2235–2244.

[31] A. Ehtesham, A. Singh, G. K. Gupta, and S. Kumar, "A survey of agent interoperability protocols: Model context protocol (mcp), agent communication protocol (acp), agent-to-agent protocol (a2a), and agent network protocol (anp)," *arXiv preprint arXiv:2505.02279*, 2025.

[32] V. Tupe and S. Thube, "Demonstrating multi-agent collaboration via agent-to-agent and model context protocols: An it incident response case study," in *2025 IEEE International Conference on Service-Oriented System Engineering (SOSE)*. IEEE, 2025, pp. 1–5.

[33] J. Cui, Z. Li, L. Xing, and X. Liao, "Safeguard-by-development: A privacy-enhanced development paradigm for multi-agent collaboration systems," *arXiv preprint arXiv:2505.04799*, 2025.

[34] L. Beckenbauer, J.-L. Loewe, G. Zheng, and A. Brintrup, "Orchestrator: Active inference for multi-agent systems in long-horizon tasks," *arXiv preprint arXiv:2509.05651*, 2025.

[35] X. Li, S. Wang, S. Zeng, Y. Wu, and Y. Yang, "A survey on llm-based multi-agent systems: workflow, infrastructure, and challenges," *Vicinagearth*, vol. 1, no. 1, p. 9, 2024.

[36] J. Lin, J.-J. Gehrt, R. Zweigel, and D. Abel, "Cooperative localization of networked multi-agent system," in *Proceedings of the 32nd International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2019)*, 2019, pp. 1976–1990.

[37] P. Natarajan, P. K. Atrey, and M. Kankanhalli, "Multi-camera coordination and control in surveillance systems: A survey," *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)*, vol. 11, no. 4, pp. 1–30, 2015.

[38] K. Yang, D. Yang, J. Zhang, M. Li, Y. Liu, J. Liu, H. Wang, P. Sun, and L. Song, "Spatio-temporal domain awareness for multi-agent collaborative perception," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 23 383–23 392.

[39] S. Hong, Y. Liu, Z. Li, S. Li, and Y. He, "Multi-agent collaborative perception via motion-aware robust communication network," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 15 301–15 310.

[40] A. Singh, G. Raut, and A. Choudhary, "Multi-agent collaborative perception for robotic fleet: A systematic review," in *European Conference on Computer Vision*. Springer, 2024, pp. 1–15.

[41] T. M. Cao, H. A. Pham, M. Walter, V. Gies, and T. Soriano, "Multi-agent robot swarms: A review of sensing and perceptual strategies for robocup soccer," in *2025 11th International Conference on Mechatronics and Robotics Engineering (ICMRE)*. IEEE, 2025, pp. 126–131.

[42] Y. Jia, Y. Song, B. Xiong, J. Cheng, W. Zhang, S. X. Yang, and S. Kwong, "Hierarchical perception-improving for decentralized multi-robot motion planning in complex scenarios," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 7, pp. 6486–6500, 2024.

[43] H. Zhang, Z. Wang, Q. Lyu, Z. Zhang, S. Chen, T. Shu, B. Dariush, K. Lee, Y. Du, and C. Gan, "Combo: compositional world models for embodied multi-agent cooperation," *arXiv preprint arXiv:2404.10775*, 2024.

[44] S. Hong, M. Zhuge, J. Chen, X. Zheng, Y. Cheng, J. Wang, C. Zhang, Z. Wang, S. K. S. Yau, Z. Lin *et al.*, "Metagpt: Meta programming for a multi-agent collaborative framework," in *The Twelfth International Conference on Learning Representations*, 2023.

[45] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, S. Zhang, J. Liu *et al.*, "Autogen: Enabling next-gen llm applications via multi-agent conversations," in *First Conference on Language Modeling*, 2024.

[46] S. Song, Y. Lin, S. Han, C. Yao, H. Wu, S. Wang, and K. Lv, "Code: Communication delay-tolerant multi-agent collaboration via dual alignment of intent and timeliness," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 22, 2025, pp. 23 304–23 312.

[47] N. Wang, Y. Zhang, S. Liu, X. Zhao, G. Cheng, and Y. Xu, "Drama: A dynamic and robust allocation-based multi-agent system for changing environments," *arXiv preprint arXiv:2508.04332*, 2025.

[48] J. He, C. Treude, and D. Lo, "Llm-based multi-agent systems for software engineering: Literature review, vision, and the road ahead," *ACM Transactions on Software Engineering and Methodology*, vol. 34, no. 5, pp. 1–30, 2025.

[49] H. Zhang, W. Du, J. Shan, Q. Zhou, Y. Du, J. B. Tenenbaum, T. Shu, and C. Gan, "Building cooperative embodied agents modularly with large language models," *arXiv preprint arXiv:2307.02485*, 2023.

[50] C. Zhang, K. Yang, S. Hu, Z. Wang, G. Li, Y. Sun, C. Zhang, Z. Zhang, A. Liu, S.-C. Zhu *et al.*, "Proagent: building proactive cooperative agents with large language models," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 16, 2024, pp. 17 591–17 599.

[51] Y. Cai, S. Mao, W. Wu, Z. Wang, Y. Liang, T. Ge, C. Wu, W. You,

T. Song, Y. Xia *et al.*, "Low-code llm: Visual programming over llms," *arXiv preprint arXiv:2304.08103*, vol. 2, p. 1, 2023.

[52] Y. Ishibashi and Y. Nishimura, "Self-organized agents: A llm multi-agent framework toward ultra large-scale code generation and optimization," *arXiv preprint arXiv:2404.02183*, 2024.

[53] M. Josifoski, L. Klein, M. Peyrard, N. Baldwin, Y. Li, S. Geng, J. P. Schnitzler, Y. Yao, J. Wei, D. Paul *et al.*, "Flows: Building blocks of reasoning and collaborating ai," *arXiv preprint arXiv:2308.01285*, 2023.

[54] G. Li, H. Hammoud, H. Itani, D. Khizbullin, and B. Ghanem, "Camel: Communicative agents for" mind" exploration of large language model society," *Advances in Neural Information Processing Systems*, vol. 36, pp. 51 991–52 008, 2023.

[55] W. Chen, Y. Su, J. Zuo, C. Yang, C. Yuan, C. Qian, C.-M. Chan, Y. Qin, Y. Lu, R. Xie *et al.*, "Agentverse: Facilitating multi-agent collaboration and exploring emergent behaviors in agents," *arXiv preprint arXiv:2308.10848*, vol. 2, no. 4, p. 6, 2023.

[56] B. Gan, Y. Zhao, T. Zhang, J. Huang, Y. Li, S. X. Teo, C. Zhang, and W. Shi, "Master: A multi-agent system with llm specialized mcts," *arXiv preprint arXiv:2501.14304*, 2025.

[57] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," *International journal of computer vision*, vol. 129, no. 6, pp. 1789–1819, 2021.

[58] J. H. Cho and B. Hariharan, "On the efficacy of knowledge distillation," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 4794–4802.

[59] Y. Tang, W. Chen, Y. Luo, and Y. Zhang, "Humble teachers teach better students for semi-supervised object detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 3132–3141.

[60] P. Mi, J. Lin, Y. Zhou, Y. Shen, G. Luo, X. Sun, L. Cao, R. Fu, Q. Xu, and R. Ji, "Active teacher for semi-supervised object detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 14 482–14 491.

[61] J. Na, J.-W. Ha, H. J. Chang, D. Han, and W. Hwang, "Switching temporary teachers for semi-supervised semantic segmentation," *Advances in Neural Information Processing Systems*, vol. 36, pp. 40 367–40 380, 2023.

[62] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015.

[63] K. Sohn, D. Berthelot, N. Carlini, Z. Zhang, H. Zhang, C. A. Raffel, E. D. Cubuk, A. Kurakin, and C.-L. Li, "Fixmatch: Simplifying semi-supervised learning with consistency and confidence," *Advances in neural information processing systems*, vol. 33, pp. 596–608, 2020.

[64] Z. Meng, J. Li, Y. Gong, and B.-H. Juang, "Adversarial teacher-student learning for unsupervised domain adaptation," in *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2018, pp. 5949–5953.

[65] R. Yang, T. Tian, and J. Tian, "Versatile teacher: A class-aware teacher–student framework for cross-domain adaptation," *Pattern Recognition*, vol. 158, p. 111024, 2025.

[66] J. Li, M. L. Seltzer, X. Wang, R. Zhao, and Y. Gong, "Large-scale domain adaptation via teacher-student learning," *arXiv preprint arXiv:1708.05466*, 2017.

[67] D. Anand, V. Gupta, P. Paruchuri, and B. Ravindran, "An enhanced advising model in teacher-student framework using state categorization," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 8, 2021, pp. 6653–6660.

[68] L. Torrey and M. Taylor, "Teaching on a budget: Agents advising agents in reinforcement learning," in *Proceedings of the 2013 international conference on Autonomous agents and multi-agent systems*, 2013, pp. 1053–1060.

[69] F. Silva, R. Glatt, and A. Costa, "An advising framework for multiagent reinforcement learning systems," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 31, no. 1, 2017.

[70] X. Li, P. Tang, F. Chen, and X. Liang, "Knowledge distillation for large language models based on global keywords and chain of thought," in *CCF International Conference on Natural Language Processing and Chinese Computing*. Springer, 2025, pp. 16–28.

[71] X. Chen, S. Zhou, K. Liang, X. Sun, and X. Liu, "Skip-thinking: Chunk-wise chain-of-thought distillation enable smaller language models to reason better and faster," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 12 153–12 168.

[72] T. M. Cover, *Elements of information theory*. John Wiley & Sons, 1999.

[73] G. Hu, Y. Zhu, D. Zhao, M. Zhao, and J. Hao, "Event-triggered communication network with limited-bandwidth constraint for multi-agent reinforcement learning," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 8, pp. 3966–3978, 2021.

[74] L. Yu, Q. Wang, Y. Qiu, J. Wang, X. Zhang, and Z. Han, "Effective multi-agent communication under limited bandwidth," *IEEE Transactions on Mobile Computing*, vol. 23, no. 7, pp. 7771–7784, 2023.

[75] W. Wang and Y. Lin, "Trajectory design and bandwidth assignment for uavs-enabled communication network with multi-agent deep reinforcement learning," in *2021 IEEE 94th Vehicular Technology Conference (VTC2021-Fall)*. IEEE, 2021, pp. 1–6.

[76] A. Quattrini Li, C. J. Carver, Q. Shao, X. Zhou, and S. Nelakuditi, "Communication for underwater robots: Recent trends," *Current Robotics Reports*, vol. 4, no. 2, pp. 13–22, 2023.

[77] D. Calvaresi, M. Marinoni, A. Sturm, M. Schumacher, and G. Buttazzo, "The challenge of real-time multi-agent systems for enabling iot and cps," in *Proceedings of the international conference on web intelligence*, 2017, pp. 356–364.

[78] W. Zhang, D. Yang, W. Wu, H. Peng, N. Zhang, H. Zhang, and X. Shen, "Optimizing federated learning in distributed industrial iot: A multi-agent approach," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 12, pp. 3688–3703, 2021.

[79] G. Wen, Z. Duan, W. Yu, and G. Chen, "Consensus in multi-agent systems with communication constraints," *International Journal of Robust and Nonlinear Control*, vol. 22, no. 2, pp. 170–182, 2012.

[80] N. Singh, I. Elamvazuthi, P. Nallagownden, G. Ramasamy, and A. Jangra, "Communication latency in multi-agent system for microgrid," in *2019 IEEE Student Conference on Research and Development (SCOReD)*. IEEE, 2019, pp. 300–305.

[81] T. Yuan, H.-M. Chung, J. Yuan, and X. Fu, "Dacom: Learning delay-aware communication for multi-agent reinforcement learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 10, 2023, pp. 11 763–11 771.

[82] S. Guo and X. Zhao, "Multi-agent deep reinforcement learning based transmission latency minimization for delay-sensitive cognitive satellite-uav networks," *IEEE Transactions on Communications*, vol. 71, no. 1, pp. 131–144, 2022.

[83] M. Stojanovic, "Underwater acoustic communications: Design considerations on the physical layer," in *2008 Fifth Annual Conference on Wireless on Demand Network Systems and Services*. IEEE, 2008, pp. 1–10.

[84] T. Zeeshan, A. Kumar, S. Pirttikangas, and S. Tarkoma, "Large language model based multi-agent system augmented complex event processing pipeline for internet of multimedia things," *arXiv preprint arXiv:2501.00906*, 2025.

[85] T. Yang, P. Feng, Q. Guo, J. Zhang, J. Ning, X. Wang, and Z. Mao, "Autohma-llm: Efficient task coordination and execution in heterogeneous multi-agent systems using hybrid large language models," *IEEE Transactions on Cognitive Communications and Networking*, 2025.

[86] J. Chen, J. Shi, Q. Chen, and M. Guo, "Kairos: Low-latency multi-agent serving with shared llms and excessive loads in the public cloud," *arXiv preprint arXiv:2508.06948*, 2025.

[87] K. Saleem, L. Wang, S. Bharany, K. Ouahada, A. U. Rehman, and H. Hamam, "Intelligent multi-agent model for energy-efficient

communication in wireless sensor networks," *EURASIP Journal on Information Security*, vol. 2024, no. 1, p. 9, 2024.

[88] G. Hu, Y. Zhu, D. Zhao, M. Zhao, and J. Hao, "Event-triggered multi-agent reinforcement learning with communication under limited-bandwidth constraint," *arXiv preprint arXiv:2010.04978*, 2020.

[89] D. Jarne Ornia and M. Mazo Jr, "Robust event-driven interactions in cooperative multi-agent learning," in *International Conference on Formal Modeling and Analysis of Timed Systems*. Springer, 2022, pp. 281–297.

[90] V. Erofeeva, O. Granichin, V. Pankov, and Z. Volkovich, "Communication-efficient decentralized clustering for dynamical multi-agent systems," *PLoS One*, vol. 20, no. 7, p. e0327396, 2025.

[91] Y. S. Nasir and D. Guo, "Multi-agent deep reinforcement learning for dynamic power allocation in wireless networks," *IEEE Journal on selected areas in communications*, vol. 37, no. 10, pp. 2239–2250, 2019.

[92] J. Foerster, I. A. Assael, N. De Freitas, and S. Whiteson, "Learning to communicate with deep multi-agent reinforcement learning," *Advances in neural information processing systems*, vol. 29, 2016.

[93] C. Zhang and V. Lesser, "Coordinating multi-agent reinforcement learning with limited communication," in *Proceedings of the 2013 international conference on Autonomous agents and multi-agent systems*, 2013, pp. 1101–1108.

[94] A. Luvisutto, A. Celani, F. Renda, C. Stefanini, and G. De Masi, "Enhancing collaboration in uncertain environment: Multi-agent reinforcement learning for underwater monitoring," *Expert Systems with Applications*, vol. 277, p. 127256, 2025.

[95] A. Mainwaring, D. Culler, J. Polastre, R. Szewczyk, and J. Anderson, "Wireless sensor networks for habitat monitoring," in *Proceedings of the 1st ACM international workshop on Wireless sensor networks and applications*, 2002, pp. 88–97.

[96] X. Chen, C. Gao, C. Chen, G. Zhang, and Y. Liu, "An empirical study on challenges for llm application developers," *ACM Transactions on Software Engineering and Methodology*, 2025.

[97] P. P. Ray, "A survey on model context protocol: Architecture, state-of-the-art, challenges and future directions," *Authorea Preprints*, 2025.

[98] B. Yan, Z. Zhou, L. Zhang, L. Zhang, Z. Zhou, D. Miao, Z. Li, C. Li, and X. Zhang, "Beyond self-talk: A communication-centric survey of llm-based multi-agent systems," *arXiv preprint arXiv:2502.14321*, 2025.

[99] I. Habler, K. Huang, V. S. Narajala, and P. Kulkarni, "Building a secure agentic ai application leveraging a2a protocol," *arXiv preprint arXiv:2504.16902*, 2025.

[100] P. P. Ray, "A review on agent-to-agent protocol: Concept, state-of-the-art, challenges and future directions," *Authorea Preprints*, 2025.

[101] K. Huang and C. Hughes, "Agentic ai communication protocols and security," in *Securing AI Agents: Foundations, Frameworks, and Real-World Deployment*. Springer, 2025, pp. 81–110.

[102] C. Bühler, M. Biagiola, L. Di Grazia, and G. Salvaneschi, "Securing ai agent execution," *arXiv preprint arXiv:2510.21236*, 2025.

[103] F. Christianos, G. Papoudakis, M. A. Rahman, and S. V. Albrecht, "Scaling multi-agent reinforcement learning with selective parameter sharing," in *International Conference on Machine Learning*. PMLR, 2021, pp. 1989–1998.

[104] J. K. Terry, N. Grammel, S. Son, B. Black, and A. Agrawal, "Revisiting parameter sharing in multi-agent deep reinforcement learning," 2023. [Online]. Available: https://arxiv.org/abs/2005.13625

[105] J. Zou, X. Yang, R. Qiu, G. Li, K. Tieu, P. Lu, K. Shen, H. Tong, Y. Choi, J. He, J. Zou, M. Wang, and L. Yang, "Latent collaboration in multi-agent systems," 2025.

[106] X. Liu and Y. Tan, "Feudal latent space exploration for coordinated multi-agent reinforcement learning," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 10, pp. 7775–7783, 2023.

[107] C. You, Y. Wu, J. Cai, Q. Luo, and Y. Zhou, "Dynamic subtask representation and assignment in cooperative multi-agent tasks," *Neurocomputing*, vol. 628, p. 129535, 2025.

[108] H. Ye, Z. Gao, M. Ma, Q. Wang, Y. Fu, M.-Y. Chung, Y. Lin, Z. Liu, J. Zhang, D. Zhuo *et al.*, "Kvcomm: Online cross-context kv-cache communication for efficient llm-based multi-agent systems," *arXiv preprint arXiv:2510.12872*, 2025.

[109] Y. Li, Y. Du, J. Zhang, L. Hou, P. Grabowski, Y. Li, and E. Ie, "Improving multi-agent debate with sparse communication topology," *arXiv preprint arXiv:2406.11776*, 2024.

[110] T. Liu, X. Wang, W. Huang, W. Xu, Y. Zeng, L. Jiang, H. Yang, and J. Li, "Groupdebate: Enhancing the efficiency of multi-agent debate using group discussion," *arXiv preprint arXiv:2409.14051*, 2024.

[111] W. Chen, J. Yuan, C. Qian, C. Yang, Z. Liu, and M. Sun, "Optima: Optimizing effectiveness and efficiency for llm-based multi-agent system," in *Findings of the Association for Computational Linguistics: ACL 2025*, 2025, pp. 11 534–11 557.

[112] M. Rafiee and A. M. Bayen, "Optimal network topology design in multi-agent systems for efficient average consensus," in *49th IEEE Conference on Decision and Control (CDC)*. IEEE, 2010, pp. 3877–3883.

[113] M. R. B. Khan, R. Jidin, and J. Pasupuleti, "Multi-agent based distributed control architecture for microgrid energy management and optimization," *Energy Conversion and Management*, vol. 112, pp. 288–307, 2016.

[114] G. Zhang, Y. Yue, X. Sun, G. Wan, M. Yu, J. Fang, K. Wang, T. Chen, and D. Cheng, "G-designer: Architecting multi-agent communication topologies via graph neural networks," *arXiv preprint arXiv:2410.11782*, 2024.

[115] Z. Wang, Y. Wang, X. Liu, L. Ding, M. Zhang, J. Liu, and M. Zhang, "Agentdropout: Dynamic agent elimination for token-efficient and high-performance llm-based multi-agent collaboration," *arXiv preprint arXiv:2503.18891*, 2025.

[116] G. Zhang, Y. Yue, Z. Li, S. Yun, G. Wan, K. Wang, D. Cheng, J. X. Yu, and T. Chen, "Cut the crap: An economical communication pipeline for llm-based multi-agent systems," *arXiv preprint arXiv:2410.02506*, 2024.

[117] S. Li, Y. Liu, Q. Wen, C. Zhang, and S. Pan, "Assemble your crew: Automatic multi-agent communication topology design via autoregressive graph generation," *arXiv preprint arXiv:2507.18224*, 2025.

[118] S. Eo, H. Moon, E. H. Zi, C. Park, and H. Lim, "Debate only when necessary: Adaptive multiagent collaboration for efficient llm reasoning," *arXiv preprint arXiv:2504.05047*, 2025.

[119] H. Zhang, Y. Shi, X. Gu, Z. Zhang, H. You, L. Gan, Y. Yuan, and J. Huang, "Hyperagent: Leveraging hypergraphs for topology optimization in multi-agent communication," *arXiv preprint arXiv:2510.10611*, 2025.

[120] Y. Xia, D. Dittler, N. Jazdi, H. Chen, and M. Weyrich, "Llm experiments with simulation: Large language model multi-agent system for simulation model parametrization in digital twins," in *2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA)*. IEEE, 2024, pp. 1–4.

[121] Y. Yang, H. Chai, S. Shao, Y. Song, S. Qi, R. Rui, and W. Zhang, "Agentnet: Decentralized evolutionary coordination for llm-based multi-agent systems," *arXiv preprint arXiv:2504.00587*, 2025.

[122] Y. Quan, R. Qi, Y. Ni, Z. Li, X. Xu, K. Huang, and X. Guo, "The moderator mechanism: A novel design in distributed multi-agent large language models," in *2025 IEEE 2nd International Conference on Electronics, Communications and Intelligent Science (ECIS)*. IEEE, 2025, pp. 1–7.

[123] S. Agashe, Y. Fan, A. Reyna, and X. E. Wang, "Llm-coordination: evaluating and analyzing multi-agent coordination abilities in large language models," in *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025, pp. 8038–8057.

[124] A. Ghafarollahi and M. J. Buehler, "Sciagents: automating scientific discovery through bioinspired multi-agent intelligent graph reasoning," *Advanced Materials*, vol. 37, no. 22, p. 2413523, 2025.

[125] Z. Li, Y. Hu, and W. Wang, "Encouraging good processes without the need for good answers: Reinforcement learning for llm agent planning," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 2025, pp. 1654–1666.

[126] A. Smit, P. Duckworth, N. Grinsztajn, T. D. Barrett, and A. Pretorius, "Should we be going mad? a look at multi-agent debate strategies for llms," *arXiv preprint arXiv:2311.17371*, 2023.

[127] Q. Wang, Z. Tang, Z. Jiang, N. Chen, T. Wang, and B. He, "Agenttaxo: Dissecting and benchmarking token distribution of llm multi-agent systems," in *ICLR 2025 Workshop on Foundation Models in the Wild*, 2025.

[128] B. Yang, X. He, H. Gao, Y. Cao, X. Li, and D. Hsu, "Codeagents: A token-efficient framework for codified multi-agent reasoning in llms," *arXiv preprint arXiv:2507.03254*, 2025.

[129] Y. Zhang, S. Yang, C. Bai, F. Wu, X. Li, Z. Wang, and X. Li, "Towards efficient llm grounding for embodied multi-agent collaboration," in *Findings of the Association for Computational Linguistics: ACL 2025*, 2025, pp. 1663–1699.

[130] Y. Liu, Y. Liu, X. Zhang, X. Chen, and R. Yan, "The truth becomes clearer through debate! multi-agent systems with large language models unmask fake news," in *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2025, pp. 504–514.

[131] Y. Wu, J. Guo, D. Li, H. P. Zou, W.-C. Huang, Y. Chen, Z. Wang, W. Zhang, Y. Li, M. Zhang *et al.*, "Psg-agent: Personality-aware safety guardrail for llm-based agents," *arXiv preprint arXiv:2509.23614*, 2025.

[132] K. Zhu, J. Zhang, Z. Qi, N. Shang, Z. Liu, P. Han, Y. Su, H. Yu, and J. You, "Safescientist: Toward risk-aware scientific discoveries by llm agents," *arXiv preprint arXiv:2505.23559*, 2025.

[133] Z. Xu, M. Qi, S. Wu, L. Zhang, Q. Wei, H. He, and N. Li, "The trust paradox in llm-based multi-agent systems: When collaboration becomes a security vulnerability," *arXiv preprint arXiv:2510.18563*, 2025.

[134] Z. Lin, J. Lan, C. Anagnostopoulos, Z. Tian, and D. Flynn, "Safety-critical multi-agent mcts for mixed traffic coordination at unsignalized intersections," *IEEE Transactions on Intelligent Transportation Systems*, 2025.

[135] L. Wang, Y. Zhang, Y. Hu, W. Wang, C. Zhang, Y. Gao, J. Hao, T. Lv, and C. Fan, "Individual reward assisted multi-agent reinforcement learning," in *International conference on machine learning*. PMLR, 2022, pp. 23 417–23 432.

[136] L. Canese, G. C. Cardarilli, L. Di Nunzio, R. Fazzolari, D. Giardino, M. Re, and S. Spanò, "Multi-agent reinforcement learning: A review of challenges and applications," *Applied Sciences*, vol. 11, no. 11, p. 4948, 2021.

[137] J. G. Kuba, M. Wen, L. Meng, H. Zhang, D. Mguni, J. Wang, Y. Yang *et al.*, "Settling the variance of multi-agent policy gradients," *Advances in Neural Information Processing Systems*, vol. 34, pp. 13 458–13 470, 2021.

[138] J. Shao, Y. Qu, C. Chen, H. Zhang, and X. Ji, "Counterfactual conservative q learning for offline multi-agent reinforcement learning," *Advances in Neural Information Processing Systems*, vol. 36, pp. 77 290–77 312, 2023.

[139] T. Mesnard, T. Weber, F. Viola, S. Thakoor, A. Saade, A. Harutyunyan, W. Dabney, T. Stepleton, N. Heess, A. Guez *et al.*, "Counterfactual credit assignment in model-free reinforcement learning," *arXiv preprint arXiv:2011.09464*, 2020.

[140] R. Zhu, X. Cheng, K. Liu, B. Zhu, D. Jin, N. Parihar, Z. Xu, and O. Gao, "Sheetmind: An end-to-end llm-powered multi-agent framework for spreadsheet automation," *arXiv preprint arXiv:2506.12339*, 2025.

[141] J. Zhang, K. Cai, Y. Fan, N. Liu, and K. Wang, "Mat-agent: Adaptive multi-agent training optimization," *arXiv preprint arXiv:2510.17845*, 2025.

[142] J. Wang, J. Wiens, and S. Lundberg, "Shapley flow: A graph-based approach to interpreting model predictions," in *International Conference on Artificial Intelligence and Statistics*. PMLR, 2021, pp. 721–729.

[143] J. Li, K. Kuang, B. Wang, F. Liu, L. Chen, F. Wu, and J. Xiao, "Shapley counterfactual credits for multi-agent reinforcement learning," in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, 2021, pp. 934–942.

[144] J. Wang, Y. Zhang, T.-K. Kim, and Y. Gu, "Shapley q-value: A local reward approach to solve global reward games," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 34, no. 05, 2020, pp. 7285–7292.

[145] Y. Yang, B. Huang, S. Qi, C. Feng, H. Hu, Y. Zhu, J. Hu, H. Zhao, Z. He, X. Liu *et al.*, "Who's the mvp? a game-theoretic evaluation benchmark for modular attribution in llm agents," *arXiv preprint arXiv:2502.00510*, 2025.

[146] Y. Xia, T. Wang, S. Zhang, Z. Weng, B. Cao, and S. C. Liew, "Hivemind: Contribution-guided online prompt optimization of llm multi-agent systems," *arXiv preprint arXiv:2512.06432*, 2025.

[147] M. Horovicz, "Agentshap: Interpreting llm agent tool importance with monte carlo shapley value estimation," *arXiv preprint arXiv:2512.12597*, 2025.

[148] Y. Hua, H. Chen, S. Wang, W. Li, X. Wang, and J. Luo, "Shapley-coop: Credit assignment for emergent cooperation in self-interested llm agents," *arXiv preprint arXiv:2506.07388*, 2025.

[149] F. Christianos, L. Schäfer, and S. Albrecht, "Shared experience actor-critic for multi-agent reinforcement learning," *Advances in neural information processing systems*, vol. 33, pp. 10 707–10 717, 2020.

[150] X. Xu, "Symmetry-driven ctde: Enhancing scalability and sample efficiency in marl," in *2025 10th International Conference on Intelligent Computing and Signal Processing (ICSP)*. IEEE, 2025, pp. 744–748.

[151] Y. Zhou, S. Liu, Y. Qing, K. Chen, T. Zheng, Y. Huang, J. Song, and M. Song, "Is centralized training with decentralized execution framework centralized enough for marl?" *arXiv preprint arXiv:2305.17352*, 2023.

[152] Y. Chen, Y. Wang, S. Zhu, H. Yu, T. Feng, M. Zhang, M. Patwary, and J. You, "Multi-agent evolve: Llm self-improve through co-evolution," *arXiv preprint arXiv:2510.23595*, 2025.

[153] O. Ozer, G. Wu, Y. Wang, D. Dosti, H. Zhang, and V. De La Rue, "Mar: Multi-agent reflexion improves reasoning abilities in llms," *arXiv preprint arXiv:2512.20845*, 2025.

[154] W. Wang, I. Obi, and B.-C. Min, "Multi-agent llm actor-critic framework for social robot navigation," *arXiv preprint arXiv:2503.09758*, 2025.

[155] Y. Yue, G. Zhang, B. Liu, G. Wan, K. Wang, D. Cheng, and Y. Qi, "Masrouter: Learning to route llms for multi-agent systems," *arXiv preprint arXiv:2502.11133*, 2025.

[156] Y. Li, J. H. Mohamud, C. Sun, D. Wu, and B. Boulet, "Leveraging llms as meta-judges: A multi-agent framework for evaluating llm judgments," *arXiv preprint arXiv:2504.17087*, 2025.

[157] P. S. Trakas, A. Tantoulas, and C. P. Bechlioulis, "Formation control of nonlinear multi-agent systems with nested input saturation," *Applied Sciences*, vol. 14, no. 1, p. 213, 2023.

[158] M. Choudhury, S. Saisubramanian, H. Zhang, and S. Zilberstein, "Minimizing negative side effects in cooperative multi-agent systems using distributed coordination," in *The International FLAIRS Conference Proceedings*, vol. 37, 2024.

[159] J. Hilton, J. Tang, and J. Schulman, "Scaling laws for single-agent reinforcement learning," *arXiv preprint arXiv:2301.13442*, 2023.

[160] D. Khatri, L. Madaan, R. Tiwari, R. Bansal, S. S. Duvvuri, M. Zaheer, I. S. Dhillon, D. Brandfonbrener, and R. Agarwal,

"The art of scaling reinforcement learning compute for llms," *arXiv preprint arXiv:2510.13786*, 2025.
[161] D. Liu, F. Ren, J. Yan, G. Su, W. Gu, and S. Kato, "Scaling up multi-agent reinforcement learning: An extensive survey on scalability issues," *IEEE Access*, vol. 12, pp. 94 610–94 631, 2024.
[162] K. Zhang, Z. Yang, and T. Başar, "Multi-agent reinforcement learning: A selective overview of theories and algorithms," *Handbook of reinforcement learning and control*, pp. 321–384, 2021.
[163] J. Liao, M. Wen, J. Wang, and W. Zhang, "Marft: Multi-agent reinforcement fine-tuning," *arXiv preprint arXiv:2504.16129*, 2025.
[164] G. Bao, L. Ma, and X. Yi, "Recent advances on cooperative control of heterogeneous multi-agent systems subject to constraints: A survey," *Systems Science & Control Engineering*, vol. 10, no. 1, pp. 539–551, 2022.
[165] X. Liao, Y. Wang, Y. Han, Y. Li, C. Lin, and X. Zhu, "Heterogeneous multi-agent deep reinforcement learning for cluster-based spectrum sharing in uav swarms," *Drones*, vol. 9, no. 5, p. 377, 2025.
[166] R. M. Kazim, G. Wang, Z. Ming, J. K. Allen, and F. Mistree, "Design and deployment of swarm engineering systems: Insights, challenges, and innovations," *IEEE Access*, 2025.
[167] V. Javalera, B. Morcego, and V. Puig, "Distributed mpc for large scale systems using agent-based reinforcement learning," *IFAC Proceedings Volumes*, vol. 43, no. 8, pp. 597–602, 2010.
[168] Y. Zheng, T. Zhang, S. Li, C. Qi, Y. Zhang, and Y. Wang, "Data-driven distributed model predictive control of continuous nonlinear systems with gaussian process," *Industrial & Engineering Chemistry Research*, vol. 61, no. 49, pp. 18 187–18 202, 2022.
[169] V. Javalera, B. Morcego, and V. Puig, "Negotiation and learning in distributed mpc of large scale systems," in *Proceedings of the 2010 American Control Conference*. IEEE, 2010, pp. 3168–3173.
[170] Z. Xu and Z. Wu, "Personalized federated learning-based distributed model predictive control with predictive error compensation for nonlinear networked systems," *IEEE Transactions on Automation Science and Engineering*, 2025.
[171] A. Gorenshtein, M. Omar, B. S. Glicksberg, G. N. Nadkarni, and E. Klang, "Ai agents in clinical medicine: A systematic review," *medRxiv*, 2025.
[172] J. Chen, X. Hu, S. Liu, S. Huang, W.-W. Tu, Z. He, and L. Wen, "Llmarena: Assessing capabilities of large language models in dynamic multi-agent environments," *arXiv preprint arXiv:2402.16499*, 2024.
[173] J. Ruan, Y. Chen, B. Zhang, Z. Xu, T. Bao, G. Du, S. Shi, H. Mao, Z. Li, X. Zeng *et al.*, "Tptu: large language model-based ai agents for task planning and tool usage," *arXiv preprint arXiv:2308.03427*, 2023.
[174] J. C.-Y. Chen, S. Yun, E. Stengel-Eskin, T. Chen, and M. Bansal, "Symbolic mixture-of-experts: Adaptive skill-based routing for heterogeneous reasoning," *arXiv preprint arXiv:2503.05641*, 2025.
[175] W. Zhou, M. Mesgar, A. Friedrich, and H. Adel, "Efficient multi-agent collaboration with tool use for online planning in complex table question answering," in *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025, pp. 945–968.
[176] K. Peng, H. Qi, and T. Ma, "Graph based multi-agent reinforcement learning with evolutionary population for cooperation," *Neural Networks*, p. 108437, 2025.
[177] T. Zhu, X. Shi, X. Xu, J. Gui, and J. Cao, "Hypercomm: Hypergraph-based communication in multi-agent reinforcement learning," *Neural Networks*, vol. 178, p. 106432, 2024.
[178] T. Wu, P. Zhou, K. Liu, Y. Yuan, X. Wang, H. Huang, and D. O. Wu, "Multi-agent deep reinforcement learning for urban traffic light control in vehicular networks," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 8, pp. 8243–8256, 2020.
[179] A. Jain, J. Sridevi, U. Dabral, A. Malhotra, and I. Kapila, "Multi-agent reinforcement learning for power system operation and control," in *E3S Web of Conferences*, vol. 511. EDP Sciences, 2024, p. 01021.
[180] H. Shahbazi and F. Karbalaei, "Decentralized voltage control of power systems using multi-agent systems," *Journal of Modern Power Systems and Clean Energy*, vol. 8, no. 2, pp. 249–259, 2020.
[181] C. Gao, X. Lan, N. Li, Y. Yuan, J. Ding, Z. Zhou, F. Xu, and Y. Li, "Large language models empowered agent-based modeling and simulation: A survey and perspectives," *Humanities and Social Sciences Communications*, vol. 11, no. 1, pp. 1–24, 2024.
[182] C. Park, S. Han, X. Guo, A. E. Ozdaglar, K. Zhang, and J.-K. Kim, "Maporl: Multi-agent post-co-training for collaborative large language models with reinforcement learning," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 30 215–30 248.
[183] J. Cui, "Communication and generalization in multi-agent learning," Ph.D. dissertation, 2025.
[184] Z. Li, "Language-guided multi-agent learning in simulations: A unified framework and evaluation," *arXiv preprint arXiv:2506.04251*, 2025.
[185] W. Jia and M. Ji, "Multi-agent deep reinforcement learning for large-scale traffic signal control with spatio-temporal attention mechanism," *Applied Sciences*, vol. 15, no. 15, p. 8605, 2025.
[186] Y. Guo, B. Lin, Q. Tang, Y. Ma, H. Luo, H. Tian, and K. Chen, "Distributed traffic engineering in hybrid software defined networks: A multi-agent reinforcement learning framework," *IEEE Transactions on Network and Service Management*, 2024.
[187] A. Suzuki, M. Kobayashi, and E. Oki, "Multi-agent deep reinforcement learning for cooperative computing offloading and route optimization in multi cloud-edge networks," *IEEE Transactions on Network and Service Management*, vol. 20, no. 4, pp. 4416–4434, 2023.
[188] K. Krishnakumar, "Dynamic load balancing in cloud networks using multi-agent systems: Ensuring high availability and efficiency," in *2025 3rd International Conference on Communication, Security, and Artificial Intelligence (ICCSAI)*, vol. 3. IEEE, 2025, pp. 872–877.
[189] Q. Huang, J. Vora, P. Liang, and J. Leskovec, "Mlagentbench: Evaluating language agents on machine learning experimentation," *arXiv preprint arXiv:2310.03302*, 2023.
[190] L. Zhang, H. Wang, L. Cheng, L. Deng, and T. Ward, "Adversarial testing in llms: Insights into decision-making vulnerabilities," *arXiv preprint arXiv:2505.13195*, 2025.
[191] T. I. U. L.-H. SCENARIOS, "Ultrahorizon: Benchmarking agent capabili-ties in ultra long-horizon scenarios."
[192] Z. Qiao, G. Chen, X. Chen, D. Yu, W. Yin, X. Wang, Z. Zhang, B. Li, H. Yin, K. Li *et al.*, "Webresearcher: Unleashing unbounded reasoning capability in long-horizon agents," *arXiv preprint arXiv:2509.13309*, 2025.
[193] S. Mallampati, R. Shelim, W. Saad, and N. Ramakrishnan, "Dynamic strategy adaptation in multi-agent environments with large language models," *arXiv preprint arXiv:2507.02002*, 2025.
[194] J. Zheng, C. Shi, X. Cai, Q. Li, D. Zhang, C. Li, D. Yu, and Q. Ma, "Lifelong learning of large language model based agents: A roadmap," *arXiv preprint arXiv:2501.07278*, 2025.
[195] J.-t. Huang, J. Zhou, T. Jin, X. Zhou, Z. Chen, W. Wang, Y. Yuan, M. R. Lyu, and M. Sap, "On the resilience of llm-based multi-agent collaboration with faulty agents," *arXiv preprint arXiv:2408.00989*, 2024.
[196] R. Ghosh, "Multi-agent memento: A theoretically grounded framework for coordinated memory-augmented continuous learning with formal convergence guarantees in high-stakes financial applications," *Authorea Preprints*, 2025.
[197] J. Xiao, G. Yuan, J. He, K. Fang, and Z. Wang, "Graph attention mechanism based reinforcement learning for multi-agent flocking

control in communication-restricted environment," *Information Sciences*, vol. 620, pp. 142–157, 2023.

[198] G. Liu, C. Somarakis, and N. Motee, "Emergence of cascading risk and role of spatial locations of collisions in time-delayed platoon of vehicles," in *2022 IEEE 61st Conference on Decision and Control (CDC)*. IEEE, 2022, pp. 6460–6465.

[199] J. Hu, Y. Dong, Z. Ding, and X. Huang, "Enhancing robustness of llm-driven multi-agent systems through randomized smoothing," *arXiv preprint arXiv:2507.04105*, 2025.

[200] B. Liu, Z. Pu, Y. Pan, J. Yi, Y. Liang, and D. Zhang, "Lazy agents: A new perspective on solving sparse reward problem in multi-agent reinforcement learning," in *International Conference on Machine Learning*. PMLR, 2023, pp. 21 937–21 950.

[201] S. Sun, H. Liu, K. Xu, and B. Ding, "Leaders and collaborators: Addressing sparse reward challenges in multi-agent reinforcement learning," *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2024.

[202] S. Choudhury, J. K. Gupta, P. Morales, and M. J. Kochenderfer, "Scalable online planning for multi-agent mdps," *Journal of Artificial Intelligence Research*, vol. 73, pp. 821–846, 2022.

[203] R. de Kock, A. Pretorius, and J. Shock, "Is an exponentially growing action space really that bad? validating a core assumption for using multi-agent rl," in *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems*, 2025, pp. 2490–2492.

[204] H. Singh, R. J. Das, M. Han, P. Nakov, and I. Laptev, "Malmm: Multi-agent large language models for zero-shot robotics manipulation," *arXiv preprint arXiv:2411.17636*, 2024.

[205] M. Zhuge, W. Wang, L. Kirsch, F. Faccio, D. Khizbullin, and J. Schmidhuber, "Gptswarm: Language agents as optimizable graphs," in *Forty-first International Conference on Machine Learning*, 2024.

[206] T. Ma, K. Peng, H. Rong, Y. Qian, and N. Al-Nabhan, "Hierarchical coordination multi-agent reinforcement learning with spatio-temporal abstraction," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 8, no. 1, pp. 533–547, 2023.

[207] Z. Hou, J. Tang, and Y. Wang, "Halo: Hierarchical autonomous logic-oriented orchestration for multi-agent llm systems," *arXiv preprint arXiv:2505.13516*, 2025.

[208] M. A. Hady, S. Hu, M. Pratama, Z. Cao, and R. Kowalczyk, "Multi-agent reinforcement learning for resources allocation optimization: a survey," *Artificial Intelligence Review*, vol. 58, no. 11, p. 354, 2025.

[209] B. Yan, P. Shi, C. P. Lim, Y. Sun, and R. K. Agarwal, "Security and safety-critical learning-based collaborative control for multiagent systems," *IEEE Transactions on Neural Networks and Learning Systems*, 2024.

[210] B. Li, S. Wang, Z. Guo, S. Zhu, J. Huang, J. Sun, G. Wen, and S. Wen, "Safe control framework of multi-agent systems from a performance enhancement perspective," *IEEE Transactions on Automation Science and Engineering*, 2024.

[211] Z. Xiang, L. Zheng, Y. Li, J. Hong, Q. Li, H. Xie, J. Zhang, Z. Xiong, C. Xie, C. Yang *et al.*, "Guardagent: Safeguard llm agents by a guard agent via knowledge-enabled reasoning," *arXiv preprint arXiv:2406.09187*, 2024.

[212] P. Sunehag, G. Lever, A. Gruslys, W. M. Czarnecki, V. Zambaldi, M. Jaderberg, M. Lanctot, N. Sonnerat, J. Z. Leibo, K. Tuyls *et al.*, "Value-decomposition networks for cooperative multi-agent learning," *arXiv preprint arXiv:1706.05296*, 2017.

[213] X. Du, Y. Ye, P. Zhang, Y. Yang, M. Chen, and T. Wang, "Situation-dependent causal influence-based cooperative multi-agent reinforcement learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 16, 2024, pp. 17 362–17 370.

[214] S. Shen, M. Qiu, J. Liu, W. Liu, Y. Fu, X. Liu, and C. Wang, "Resq: A residual q function-based approach for multi-agent reinforcement learning value factorization," *Advances in Neural Information Processing Systems*, vol. 35, pp. 5471–5483, 2022.

[215] W. Du, S. Ding, L. Guo, J. Zhang, C. Zhang, and L. Ding, "Value function factorization with dynamic weighting for deep multi-agent reinforcement learning," *Information Sciences*, vol. 615, pp. 191–208, 2022.

[216] W. Yang and J. Thomason, "Learning to deliberate: Meta-policy collaboration for agentic llms with multi-agent reinforcement learning," *arXiv preprint arXiv:2509.03817*, 2025.

[217] Z. Wan, Y. Li, X. Wen, Y. Song, H. Wang, L. Yang, M. Schmidt, J. Wang, W. Zhang, S. Hu *et al.*, "Rema: Learning to meta-think for llms with multi-agent reinforcement learning," *arXiv preprint arXiv:2503.09501*, 2025.

[218] G. Verma, R. Kaur, N. Srishankar, Z. Zeng, T. Balch, and M. Veloso, "Adaptagent: Adapting multimodal web agents with few-shot learning from human demonstrations," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 20 635–20 651.

[219] G. Zhang, H. Ren, C. Zhan, Z. Zhou, J. Wang, H. Zhu, W. Zhou, and S. Yan, "Memevolve: Meta-evolution of agent memory systems," *arXiv preprint arXiv:2512.18746*, 2025.

[220] M. Hu, P. Zhao, C. Xu, Q. Sun, J.-G. Lou, Q. Lin, P. Luo, and S. Rajmohan, "Agentgen: Enhancing planning abilities for large language model based agent via environment and task generation," in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1*, 2025, pp. 496–507.

[221] A. Zeng, M. Liu, R. Lu, B. Wang, X. Liu, Y. Dong, and J. Tang, "Agenttuning: Enabling generalized agent abilities for llms," in *Findings of the Association for Computational Linguistics: ACL 2024*, 2024, pp. 3053–3077.

[222] S. Tang, X. Pang, Z. Liu, B. Tang, R. Ye, T. Jin, X. Dong, Y. Wang, and S. Chen, "Synthesizing post-training data for llms through multi-agent simulation," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 23 306–23 335.

[223] V. D. Xuan, H. Vo, D. Murphy, and H. D. Nguyen, "Agentsgen: Multi-agent llm in the loop for semantic collaboration and generation of synthetic data," *arXiv preprint arXiv:2505.13466*, 2025.

[224] R. Ye, S. Tang, R. Ge, Y. Du, Z. Yin, S. Chen, and J. Shao, "Mas-gpt: Training llms to build llm-based multi-agent systems," *arXiv preprint arXiv:2503.03686*, 2025.

[225] M. Chen, "Evaluating large language models trained on code," *arXiv preprint arXiv:2107.03374*, 2021.

[226] J. Austin, A. Odena, M. Nye, M. Bosma, H. Michalewski, D. Dohan, E. Jiang, C. Cai, M. Terry, Q. Le *et al.*, "Program synthesis with large language models," *arXiv preprint arXiv:2108.07732*, 2021.

[227] Z. Mandi, S. Jain, and S. Song, "Roco: Dialectic multi-robot collaboration with large language models," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 286–299.

[228] C. Gan, J. Schwartz, S. Alter, D. Mrowca, M. Schrimpf, J. Traer, J. De Freitas, J. Kubilius, A. Bhandwaldar, N. Haber *et al.*, "Threedworld: A platform for interactive multi-modal physical simulation," *arXiv preprint arXiv:2007.04954*, 2020.

[229] S. C. Duncan, "Minecraft, beyond construction and survival," 2011.

[230] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, "Measuring massive multitask language understanding," *arXiv preprint arXiv:2009.03300*, 2020.

[231] D. Jin, E. Pan, N. Oufattole, W.-H. Weng, H. Fang, and P. Szolovits, "What disease does this patient have? a large-scale open domain question answering dataset from medical exams," *Applied Sciences*, vol. 11, no. 14, p. 6421, 2021.

[232] Q. Jin, B. Dhingra, Z. Liu, W. Cohen, and X. Lu, "Pubmedqa: A dataset for biomedical research question answering," in *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint*

*conference on natural language processing (EMNLP-IJCNLP)*, 2019, pp. 2567–2577.

[233] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano *et al.*, "Training verifiers to solve math word problems," *arXiv preprint arXiv:2110.14168*, 2021.

[234] M. Geva, D. Khashabi, E. Segal, T. Khot, D. Roth, and J. Berant, "Did aristotle use a laptop? a question answering benchmark with implicit reasoning strategies," *Transactions of the Association for Computational Linguistics*, vol. 9, pp. 346–361, 2021.

[235] M. A. Islam, M. E. Ali, and M. R. Parvez, "Mapcoder: Multi-agent code generation for competitive problem solving," *arXiv preprint arXiv:2405.11403*, 2024.

[236] G. Zhang, L. Niu, J. Fang, K. Wang, L. Bai, and X. Wang, "Multi-agent architecture search via agentic supernet," *arXiv preprint arXiv:2502.04180*, 2025.

[237] I. Abbasnejad, X. Liu, and A. Roy, "Deciding the path: Leveraging multi-agent systems for solving complex tasks," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 4216–4225.

[238] A. Deo, "Qaagent: A multiagent system for unit test generation via natural language pseudocode (student abstract)," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 28, 2025, pp. 29 345–29 347.

[239] H. Li, Y. Shi, S. Lin, X. Gu, H. Lian, X. Wang, Y. Jia, T. Huang, and Q. Wang, "Swe-debate: Competitive multi-agent debate for software issue resolution," *arXiv preprint arXiv:2507.23348*, 2025.

[240] D. Chen, S. Lin, M. Zeng, D. Zan, J.-G. Wang, A. Cheshkov, J. Sun, H. Yu, G. Dong, A. Aliev *et al.*, "Coder: Issue resolving with multi-agent and task graphs," *arXiv preprint arXiv:2406.01304*, 2024.

[241] W. Chen, Z. You, R. Li, Y. Guan, C. Qian, C. Zhao, C. Yang, R. Xie, Z. Liu, and M. Sun, "Internet of agents: Weaving a web of heterogeneous agents for collaborative intelligence," *arXiv preprint arXiv:2407.07061*, 2024.

[242] J. Liu, P. Zhou, Y. Du, A.-H. Tan, C. G. Snoek, J.-J. Sonke, and E. Gavves, "Capo: Cooperative plan optimization for efficient embodied multi-agent cooperation," *arXiv preprint arXiv:2411.04679*, 2024.

[243] L. Zu, L. Lin, S. Fu, N. Zhao, and P. Zhou, "Collaborative tree search for enhancing embodied multi-agent collaboration," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 29 513–29 522.

[244] S. Seo, S. Noh, J. Lee, S. Lim, W. H. Lee, and H. Kang, "Reveca: Adaptive planning and trajectory-based validation in cooperative language agents using information relevance and relative proximity," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 22, 2025, pp. 23 295–23 303.

[245] Y. Dong, X. Zhu, Z. Pan, L. Zhu, and Y. Yang, "Villager-agent: A graph-based multi-agent framework for coordinating complex task dependencies in minecraft," *arXiv preprint arXiv:2406.05720*, 2024.

[246] Q. Chai, Z. Zheng, J. Ren, D. Ye, Z. Lin, and H. Wang, "Causalmace: Causality empowered multi-agents in minecraft cooperative tasks," *arXiv preprint arXiv:2508.18797*, 2025.

[247] R. Gong, Q. Huang, X. Ma, Y. Noda, Z. Durante, Z. Zheng, D. Terzopoulos, L. Fei-Fei, J. Gao, and H. Vo, "Mindagent: Emergent gaming interaction," in *Findings of the Association for Computational Linguistics: NAACL 2024*, 2024, pp. 3154–3183.

[248] X. Tang, A. Zou, Z. Zhang, Z. Li, Y. Zhao, X. Zhang, A. Cohan, and M. Gerstein, "Medagents: Large language models as collaborators for zero-shot medical reasoning," in *Findings of the Association for Computational Linguistics: ACL 2024*, 2024, pp. 599–621.

[249] Z. Xiao, R. Zhang, Y. Feng, L. Zhu, L. Peng, and L. Yu, "A dynamic agent framework for large language model reasoning for medical and visual question answering," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 1143–1152.

[250] R. Wang, Y. Chen, W. Zhang, J. Si, H. Guan, X. Peng, and W. Lu, "Medconma: A confidence-driven multi-agent framework for medical q&a," in *Pacific-Asia Conference on Knowledge Discovery and Data Mining*. Springer, 2025, pp. 421–433.

[251] Y. Miao, J. Wen, Y. Luo, and J. Li, "Medarc: Adaptive multi-agent refinement and collaboration for enhanced medical reasoning in large language models," *International Journal of Medical Informatics*, p. 106136, 2025.

[252] K. Chen, J. Qi, J. Huo, P. Tian, F. Meng, X. Yang, and Y. Gao, "A self-evolving framework for multi-agent medical consultation based on large language models," in *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025, pp. 1–5.

[253] C. Gao, H. Jiang, D. Cai, S. Shi, and W. Lam, "Strategyllm: Large language models as strategy generators, executors, optimizers, and evaluators for problem solving," *Advances in Neural Information Processing Systems*, vol. 37, pp. 96 797–96 846, 2024.

[254] Z. He and Y. Feng, "Unleashing diverse thinking modes in llms through multi-agent collaboration," *arXiv preprint arXiv:2510.16645*, 2025.

[255] A. Aluru, M. Malik, S. Patankar, S. Kim, K. Zhu, S. O'Brien, and V. Sharma, "Smagdi: Socratic multi agent interaction graph distillation for efficient high accuracy reasoning," *arXiv preprint arXiv:2511.05528*, 2025.

[256] X. Liu, H. Yu, H. Zhang, Y. Xu, X. Lei, H. Lai, Y. Gu, H. Ding, K. Men, K. Yang *et al.*, "Agentbench: Evaluating llms as agents," *arXiv preprint arXiv:2308.03688*, 2023.

[257] M. Chang, J. Zhang, Z. Zhu, C. Yang, Y. Yang, Y. Jin, Z. Lan, L. Kong, and J. He, "Agentboard: An analytical evaluation board of multi-turn llm agents," *Advances in neural information processing systems*, vol. 37, pp. 74 325–74 362, 2024.

[258] X. Wang, Z. Wang, J. Liu, Y. Chen, L. Yuan, H. Peng, and H. Ji, "Mint: Evaluating llms in multi-turn interaction with tools and language feedback," *arXiv preprint arXiv:2309.10691*, 2023.

[259] J. Y. Koh, R. Lo, L. Jang, V. Duvvur, M. Lim, P.-Y. Huang, G. Neubig, S. Zhou, R. Salakhutdinov, and D. Fried, "Visualwebarena: Evaluating multimodal agents on realistic visual web tasks," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 881–905.

[260] Z. Chen, W. Du, W. Zhang, K. Liu, J. Liu, M. Zheng, J. Zhuo, S. Zhang, D. Lin, K. Chen *et al.*, "T-eval: Evaluating the tool utilization capability of large language models step by step," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 9510–9529.

[261] A. Zou, L. Phan, J. Wang, D. Duenas, M. Lin, M. Andriushchenko, J. Z. Kolter, M. Fredrikson, and D. Hendrycks, "Improving alignment and robustness with circuit breakers," *Advances in Neural Information Processing Systems*, vol. 37, pp. 83 345–83 373, 2024.

[262] Q. Chen, D. Pitawela, C. Zhao, G. Zhou, H.-T. Chen, and Q. Wu, "Webvln: Vision-and-language navigation on websites," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 2, 2024, pp. 1165–1173.

[263] F. Yin, Z. Wang, I.-H. Hsu, J. Yan, K. Jiang, Y. Chen, J. Gu, L. Le, K.-W. Chang, C.-Y. Lee *et al.*, "Magnet: Multi-turn tool-use data synthesis and distillation via graph translation," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 32 600–32 616.

[264] Z. Liu, T. Hoang, J. Zhang, M. Zhu, T. Lan, J. Tan, W. Yao, Z. Liu, Y. Feng, R. RN *et al.*, "Apigen: Automated pipeline for generating verifiable and diverse function-calling datasets," *Advances in Neural Information Processing Systems*, vol. 37, pp. 54 463–54 482, 2024.

[265] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Vaughan *et al.*, "The llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024.

[266] H. He, W. Yao, K. Ma, W. Yu, Y. Dai, H. Zhang, Z. Lan, and D. Yu, "Webvoyager: Building an end-to-end web agent with large multimodal models," *arXiv preprint arXiv:2401.13919*, 2024.

[267] X. Wang, Y. Chen, L. Yuan, Y. Zhang, Y. Li, H. Peng, and H. Ji, "Executable code actions elicit better llm agents," in *Forty-first International Conference on Machine Learning*, 2024.

[268] G. Cui, L. Yuan, N. Ding, G. Yao, B. He, W. Zhu, Y. Ni, G. Xie, R. Xie, Y. Lin *et al.*, "Ultrafeedback: Boosting language models with scaled ai feedback," *arXiv preprint arXiv:2310.01377*, 2023.

[269] K. He, Z. Wang, C. Zhuang, and J. Gu, "Recon-act: A self-evolving multi-agent browser-use system via web reconnaissance, tool generation, and task execution," *arXiv preprint arXiv:2509.21072*, 2025.

[270] Z. Chen, K. Liu, Q. Wang, W. Zhang, J. Liu, D. Lin, K. Chen, and F. Zhao, "Agent-flan: Designing data and methods of effective agent tuning for large language models," *arXiv preprint arXiv:2403.12881*, 2024.

[271] Y. Wang, H. Li, X. Han, P. Nakov, and T. Baldwin, "Do-not-answer: A dataset for evaluating safeguards in llms," *arXiv preprint arXiv:2308.13387*, 2023.

[272] L. Boisvert, M. Bansal, C. K. R. Evuru, G. Huang, A. Puri, A. Bose, M. Fazel, Q. Cappart, J. Stanley, A. Lacoste *et al.*, "Doomarena: A framework for testing ai agents against evolving security threats," *arXiv preprint arXiv:2504.14064*, 2025.

[273] C. Jiang, X. Pan, and M. Yang, "Think twice before you act: Enhancing agent behavioral safety with thought correction," *arXiv preprint arXiv:2505.11063*, 2025.

[274] X. Yuan, J. Li, D. Wang, Y. Chen, X. Mao, L. Huang, J. Chen, H. Xue, X. Liu, W. Wang *et al.*, "S-eval: Towards automated and comprehensive safety evaluation for large language models," *Proceedings of the ACM on Software Engineering*, vol. 2, no. ISSTA, pp. 2136–2157, 2025.

[275] J. Xu, D. Ju, M. Li, Y.-L. Boureau, J. Weston, and E. Dinan, "Bot-adversarial dialogue for safe conversational agents," in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2021, pp. 2950–2968.

[276] J. Ji, M. Liu, J. Dai, X. Pan, C. Zhang, C. Bian, B. Chen, R. Sun, Y. Wang, and Y. Yang, "Beavertails: Towards improved safety alignment of llm via a human-preference dataset," *Advances in Neural Information Processing Systems*, vol. 36, pp. 24 678–24 704, 2023.

[277] Y. Yuan, W. Jiao, W. Wang, J.-t. Huang, J. Xu, T. Liang, P. He, and Z. Tu, "Refuse whenever you feel unsafe: Improving safety in llms via decoupled refusal training," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2025, pp. 3149–3167.

[278] S. Qi, Y. Zou, P. Li, Z. Lin, X. Cheng, and D. Yu, "Amplified vulnerabilities: Structured jailbreak attacks on llm-based multi-agent debate," *arXiv preprint arXiv:2504.16489*, 2025.

[279] I. Kerboua, S. O. Shayegan, M. Thakkar, X. H. Lù, L. Boisvert, M. Caccia, J. Espinas, A. Aussem, V. Eglin, and A. Lacoste, "Focusagent: Simple yet effective ways of trimming the large context of web agents," *arXiv preprint arXiv:2510.03204*, 2025.

[280] J. Xu, D. Ju, M. Li, Y.-L. Boureau, J. Weston, and E. Dinan, "Recipes for safety in open-domain chatbots," *arXiv preprint arXiv:2010.07079*, 2020.

[281] E. Dinan, G. Abercrombie, A. S. Bergman, S. Spruit, D. Hovy, Y.-L. Boureau, and V. Rieser, "Anticipating safety issues in e2e conversational ai: Framework and tooling," *arXiv preprint arXiv:2107.03451*, 2021.

[282] M. Ung, J. Xu, and Y.-L. Boureau, "Saferdialogues: Taking feedback gracefully after conversational safety failures," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 6462–6481.

[283] D. Rosati, J. Wehner, K. Williams, L. Bartoszcze, R. Gonzales, S. Majumdar, H. Sajjad, F. Rudzicz *et al.*, "Representation noising: A defence mechanism against harmful finetuning," *Advances in Neural Information Processing Systems*, vol. 37, pp. 12 636–12 676, 2024.

[284] S. Lin, J. Hilton, and O. Evans, "Truthfulqa: Measuring how models mimic human falsehoods," in *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: long papers)*, 2022, pp. 3214–3252.

[285] N. Cointe, G. Bonnet, and O. Boissier, "Ethical judgment of agents' behaviors in multi-agent systems." in *AAMAS*, 2016, pp. 1106–1114.

[286] R. Maertens, F. M. Götz, H. F. Golino, J. Roozenbeek, C. R. Schneider, Y. Kyrychenko, J. R. Kerr, S. Stieger, W. P. McClanahan, K. Drabot *et al.*, "The misinformation susceptibility test (mist): A psychometrically validated measure of news veracity discernment," *Behavior Research Methods*, vol. 56, no. 3, pp. 1863–1899, 2024.

[287] Z. Zhang, Y. Shi, J. Zhu, W. Zhou, X. Qi, P. Zhang, and H. Li, "Trustworthy alignment of retrieval-augmented large language models via reinforcement learning," *arXiv preprint arXiv:2410.16843*, 2024.

[288] N. Joshi, J. Rando, A. Saparov, N. Kim, and H. He, "Personas as a way to model truthfulness in language models," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024, pp. 6346–6359.

[289] R. Chaput, J. Duval, O. Boissier, M. Guillermin, and S. Hassas, "A multi-agent approach to combine reasoning and learning for an ethical behavior," in *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, 2021, pp. 13–23.

[290] J. Roozenbeek, S. Van Der Linden, B. Goldberg, S. Rathje, and S. Lewandowsky, "Psychological inoculation improves resilience against misinformation on social media," *Science advances*, vol. 8, no. 34, p. eabo6254, 2022.

[291] M. Chang, G. Chhablani, A. Clegg, M. D. Cote, R. Desai, M. Hlavac, V. Karashchuk, J. Krantz, R. Mottaghi, P. Parashar *et al.*, "Partnr: A benchmark for planning and reasoning in embodied multi-agent tasks," *arXiv preprint arXiv:2411.00081*, 2024.

[292] J. Guo, Y. Chen, Y. Hao, Z. Yin, Y. Yu, and S. Li, "Towards comprehensive testing on the robustness of cooperative multi-agent reinforcement learning," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 115–122.

[293] Z. Zhu, B. Wu, Z. Zhang, and B. Wu, "Riskawarebench: Towards evaluating physical risk awareness for high-level planning of llm-based embodied agents," *arXiv e-prints*, pp. arXiv–2408, 2024.

[294] X. Zhou, H. Kim, F. Brahman, L. Jiang, H. Zhu, X. Lu, F. Xu, B. Y. Lin, Y. Choi, N. Mireshghallah *et al.*, "Haicosystem: An ecosystem for sandboxing safety risks in human-ai interactions," *arXiv preprint arXiv:2409.16427*, 2024.

[295] Y. Ruan, H. Dong, A. Wang, S. Pitis, Y. Zhou, J. Ba, Y. Dubois, C. J. Maddison, and T. Hashimoto, "Identifying the risks of lm agents with an lm-emulated sandbox," *arXiv preprint arXiv:2309.15817*, 2023.

[296] K. Zhu, J. Wang, J. Zhou, Z. Wang, H. Chen, Y. Wang, L. Yang, W. Ye, Y. Zhang, N. Zhenqiang Gong *et al.*, "Promptbench: Towards evaluating the robustness of large language models on adversarial prompts," *arXiv e-prints*, pp. arXiv–2306, 2023.

[297] T. Sadhu, A. Pesaranghader, Y. Chen, and D. H. Yi, "Athena: Safe autonomous agents with verbal contrastive learning," *arXiv preprint arXiv:2408.11021*, 2024.

[298] K. Zhu, J. Wang, Q. Zhao, R. Xu, and X. Xie, "Dynamic evaluation of large language models by meta probing agents," *arXiv preprint arXiv:2402.14865*, 2024.

[299] J. Chen, X. Huang, and Y. Li, "Dynamic supplementation of federated search results for reducing hallucinations in llms," 2024.

[300] D. Deshpande, V. Gangal, H. Mehta, J. Krishnan, A. Kannappan, and R. Qian, "Trail: Trace reasoning and agentic issue localization," *arXiv preprint arXiv:2505.08638*, 2025.

[301] T. Rebedea, R. Dinu, M. N. Sreedhar, C. Parisien, and J. Cohen, "Nemo guardrails: A toolkit for controllable and safe llm applications with programmable rails," in *Proceedings of the 2023 conference on empirical methods in natural language processing: system demonstrations*, 2023, pp. 431–445.

[302] A. Asad, S. Obadinma, R. Shayanfar, and X. Zhu, "Reddebate: Safer responses through multi-agent red teaming debates," *arXiv preprint arXiv:2506.11083*, 2025.